

ANALYSIS II FOR UNDERGRADUATE STUDENTS

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ABSTRACT. This lecture note is written to teach undergraduate students Analysis II. This lecture note is based on the book [1].

1. DIFFERENTIATION

First, review the definition of a metric space.

Definition 1.1. A set X is said to be a **metric space** if and only if there exists a function $d : X \times X \rightarrow \mathbb{R}$ such that

(a)

$$d(p, p) = 0 \quad \forall p \in X$$

(b)

$$d(p, q) > 0 \quad \text{if } p \neq q$$

(c)

$$d(p, q) = d(q, p) \quad \forall p, q \in X$$

(d) (triangle inequality)

$$d(p, q) \leq d(p, r) + d(r, q) \quad \forall p, q, r \in X.$$

Any function satisfying above four properties is called a **metric** or **distance function**. An element of a metric space X is called a **point**.

Sometime we say that (X, d) is a metric space if X is a metric space with the metric d .

Remark 1.2. Let (X, d) be an metric space. Then any subset of X is also an metric space with the metric d .

Definition 1.3. Let (X, d_X) and (Y, d_Y) be metric spaces. Assume that $E \subset X$, f is a mapping from E to Y , and p is a limit point of E . We say that f has a limit q at p and write $f(x) \rightarrow q$ as $x \rightarrow p$ or

$$\lim_{x \rightarrow p} f(x) = q$$

if and only if $q \in Y$ and for any $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$d_Y(f(x), q) < \varepsilon$$

for all $x \in E$ satisfying

$$0 < d_X(x, p) < \delta.$$

Definition 1.4. Let (X, d_X) and (Y, d_Y) be metric spaces, $E \subset X$, f is a mapping from E to Y , and $p \in E$. Then we say that f is continuous at p if and only if for any $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$d_Y(f(x), f(p)) < \varepsilon$$

for all $x \in E$ satisfying

$$d_X(x, p) < \delta.$$

Additionally, we say that f is continuous on E if and only if f is continuous at p for all $p \in E$.

Example 1.5. One of the most important examples of a metric space are the Euclidean spaces $\mathbb{R}^k$. The metric in $\mathbb{R}^k$ is defined by

$$d(x, y) = |x - y| \quad \forall x, y \in \mathbb{R}^k. \quad (1.1)$$

Recall the definition of an open interval and a closed interval in $\mathbb{R}$. We define

$$(a, b) := \{x \in \mathbb{R} : a < x < b\}$$

and

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Definition 1.6. Let a, b be real numbers such that $a < b$ and f be a real-valued function defined on the open interval (a, b) .

- (i) For $d \in \mathbb{R}$ and $c \in (a, b)$, we say that f has a **limit** d at c iff (if and only if) for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - d| < \varepsilon$$

for all $x \in (c - \delta, c) \cup (c, c + \delta)$. We use the notation

$$\lim_{x \rightarrow c} f(x) = d$$

or we say that $f(x) \rightarrow d$ as $x \rightarrow c$ iff f has a **limit** d at c .

- (ii) Similarly, for $d \in \mathbb{R}$ and $c \in [a, b)$, we say that f has a **right limit** d at c iff for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - d| < \varepsilon$$

for all $x \in (c, c + \delta)$. For the right limit, we use the notation We use the notation

$$\lim_{x \rightarrow c^+} f(x) = d$$

or

$$\lim_{x \downarrow c} f(x) = d.$$

Remark 1.7. Since $[a, b)$ is a metric space, the limit of f at a is defined in Definition 1.3. Moreover, it is easily checkable that

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a^+} f(x).$$

Exercise 1.8. Define a **left limit** of a function.

Exercise 1.9. Let f be a constant function on (a, b) , i.e. there exists a real number d such that

$$f(x) = d \quad \forall x \in (a, b).$$

Show that for all $c \in (a, b)$,

$$\lim_{x \rightarrow c} f(x) = d.$$

Exercise 1.10. Let f be a function on (a, b) and $c \in (a, b)$. Show that $\lim_{x \rightarrow c} f(x)$ exists if and only if $\lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x)$, where $\lim_{x \rightarrow c^-} f(x)$ denotes the left limit of f at c and $\lim_{x \rightarrow c^+} f(x)$ is the right limit of f at c .

Exercise 1.11. Compare Definition 1.3 and 1.6(i).

Theorem 1.12. Let f, g be real-valued functions defined on an open interval (a, b) . We define functions $f + g$ and fg on (a, b) as follows: for all $x \in (a, b)$,

$$(f + g)(x) = f(x) + g(x)$$

and

$$fg(x) = f(x)g(x).$$

Additionally, we define

$$\frac{f}{g}(x) = \frac{f(x)}{g(x)}$$

if $g(x) \neq 0$. Assume that f has a limit at c and g has a limit at c . Then

(i)

$$\lim_{x \rightarrow c} (f + g)(x) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$$

(ii)

$$\lim_{x \rightarrow c} (fg)(x) = \lim_{x \rightarrow c} f(x) \lim_{x \rightarrow c} g(x)$$

(iii)

$$\lim_{x \rightarrow c} \left(\frac{f}{g} \right)(x) = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$$

if $\lim_{x \rightarrow c} g(x) \neq 0$.

Exercise 1.13. Prove Theorem 1.12.

Exercise 1.14. Prove that formula (i), (ii), and (iii) in Theorem 1.12 hold for left limits and right limits.

Definition 1.15. Let f be a real-valued function defined on the closed interval $[a, b]$. For $c \in [a, b]$, we say that f is continuous at c iff for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon$$

for all $x \in (c - \delta, c + \delta) \cap [a, b]$. In particular, for $c \in (a, b)$, f is continuous at c iff

$$\lim_{x \rightarrow c} f(x) = f(c).$$

Exercise 1.16. (i) Define that f is **left continuous** at c and f is **right continuous** at c .

(ii) Compare that f is right continuous at a and that f is continuous at a .

(iii) Prove the two definitions 1.4 and 1.15 coincide.

Definition 1.17. Let f be a real-valued function defined on a closed interval $[a, b]$. For all $x \in [a, b]$, consider the mapping

$$\phi(t) := \frac{f(t) - f(x)}{t - x} \quad (a < t < b, \quad t \neq x).$$

Define

$$\begin{aligned} f'(x) &= \lim_{t \rightarrow x} \phi(t) \quad \text{if } x \in (a, b), \\ f'(x) &= \lim_{t \rightarrow x^+} \phi(t) \quad \text{if } x = a, \end{aligned}$$

and

$$f'(x) = \lim_{t \rightarrow x^-} \phi(t) \quad \text{if } x = b$$

provided that the limits exist, where $\lim_{t \rightarrow x^-}$ denotes the left limit and $\lim_{t \rightarrow x^+}$ is the right limit.

We can consider f' as a real-valued function whose domain is a one of the following four sets:

$$\begin{aligned} & \left\{ x \in (a, b) : \lim_{t \rightarrow x} \phi(t) \text{ exists} \right\} \\ & \left\{ x \in (a, b) : \lim_{t \rightarrow x} \phi(t) \text{ exists} \right\} \cup \{a\}, \\ & \left\{ x \in (a, b) : \lim_{t \rightarrow x} \phi(t) \text{ exists} \right\} \cup \{b\}, \\ & \left\{ x \in (a, b) : \lim_{t \rightarrow x} \phi(t) \text{ exists} \right\} \cup \{a, b\}. \end{aligned}$$

If f' is defined at x , then we say that f is **differentiable** at x and $f'(x)$ is called a **derivative** of f at x . Additionally, if f is differentiable for all $x \in E$, then we say that f is differentiable on E . We also use the notation

$$\frac{df}{dx}(x) := f'(x)$$

or

$$\frac{d}{dx}(f)(x) := f'(x)$$

to emphasize the variable.

Remark 1.18. By using Definition 1.3, we can simply define

$$f'(x) = \lim_{t \rightarrow x} \phi(t) \quad \forall x \in [a, b]$$

if the limit exists.

Exercise 1.19. (i) Define a **left derivative** and a **right derivative** of a function f on $[a, b]$.

(ii) Compare the right derivative of f at a and the derivative of f at a .

Example 1.20. For a constant $c \in \mathbb{R}$, define $f(x) = c$. Then f is differentiable on x and $f'(x) = 0$ for all $x \in \mathbb{R}$. Consider the identity function $g(x) = x$. Then g is differentiable on $\mathbb{R}$ and $g'(x) = 1$ for all $x \in \mathbb{R}$.

Theorem 1.21. Let f be a real-valued function on $[a, b]$. Assume that f is differentiable at $x \in [a, b]$. Then f is continuous at x .

Proof. Observe that for all $t, x \in [a, b]$ such that $t \neq x$,

$$f(t) - f(x) = \frac{f(t) - f(x)}{t - x} \cdot (t - x). \quad (1.2)$$

- (Case 1 : $x \in (a, b)$). Take $\lim_{t \rightarrow x}$ to both sides of (1.2). Then by Theorem 1.12, we have

$$\lim_{t \rightarrow x} (f(t) - f(x)) = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} \cdot \lim_{t \rightarrow x} (t - x) = f'(x) \cdot 0 = 0.$$

- (Case 2 : $x = a$). Use the right limit.
- (Case 3 : $x = b$). Use the left limit.

□

Exercise 1.22. Complete the proof of Theorem 1.21 for Case 2 and Case 3.

Exercise 1.23 (Difficult!). To make Theorem 1.21 reasonable, we defined “continuity” for a function defined on a closed interval in Definition 1.15. However, it is more desirable to define “continuity” for a function defined on an open interval. Explain why the definition of “continuity” for a function defined on a closed interval is bad.

Remark 1.24. It is well-known that a real-valued continuous function f on a compact set X has a maximum and minimum, i.e. there exist $x, y \in X$ such that

$$f(x) \geq f(z) \quad \forall z \in X$$

and

$$f(y) \leq f(z) \quad \forall z \in X.$$

Thus by Theorem 1.21, any real-valued differentiable function f on $[a, b]$ has a maximum and minimum.

Theorem 1.25. Let f and g be functions on $[a, b]$. Assume that both f and g are differentiable at $x \in [a, b]$. Then $f + g$ and fg are differentiable at x . If $g(x) \neq 0$, f/g is also differentiable at x . Moreover,

(i)

$$(f + g)'(x) = f'(x) + g'(x)$$

(ii)

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x)$$

(iii)

$$\left(\frac{f}{g}\right)'(x) = \frac{g(x)f'(x) - g'(x)f(x)}{g^2(x)} \quad \text{if } g(x) \neq 0,$$

where $g^2(x) = gg(x) := g(x)g(x)$. In particular,

$$\left(\frac{1}{g}\right)'(x) = -\frac{g'(x)}{g^2(x)} \quad \text{if } g(x) \neq 0. \quad (1.3)$$

Proof. Due the similarity of the proof, we only consider the case $x \in (a, b)$. First we prove (i). Since both f and g are differentiable at x ,

$$\lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} \quad \text{and} \quad \lim_{t \rightarrow x} \frac{g(t) - g(x)}{t - x}$$

exist. Thus by Theorem 1.12,

$$\lim_{t \rightarrow x} \left(\frac{f(t) - f(x)}{t - x} + \frac{g(t) - g(x)}{t - x} \right) = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} + \lim_{t \rightarrow x} \frac{g(t) - g(x)}{t - x}.$$

Next we prove (ii). Observe that for all $t \in (a, b)$ such that $t \neq x$,

$$\begin{aligned} \frac{f(t)g(t) - f(x)g(x)}{t - x} &= \frac{f(t)g(t) - f(t)g(x) + f(t)g(x) - f(x)g(x)}{t - x} \\ &= \frac{f(t)(g(t) - g(x)) + (f(t) - f(x))g(x)}{t - x} \end{aligned} \quad (1.4)$$

Due to Theorem 1.21, f and g are continuous at x and thus

$$\lim_{t \rightarrow x} f(t) = f(x) \quad \text{and} \quad \lim_{t \rightarrow x} g(t) = g(x).$$

Therefore by Theorem 1.12 and (1.4),

$$\begin{aligned} \lim_{t \rightarrow x} \frac{f(t)g(t) - f(x)g(x)}{t - x} &= \lim_{t \rightarrow x} f(t) \lim_{t \rightarrow x} \frac{g(t) - g(x)}{t - x} + g(x) \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} \\ &= f(x)g'(x) + g(x)f'(x). \end{aligned}$$

Finally we prove (iii). If we prove (1.3), then by (ii), we have

$$\begin{aligned} \left(\frac{f}{g}\right)'(x) &= \left(f \frac{1}{g}\right)'(x) = f'(x) \frac{1}{g(x)} - f(x) \frac{g'(x)}{g^2(x)} \\ &= \frac{f'(x)g(x) - f(x)g'(x)}{g^2(x)}. \end{aligned}$$

Thus it suffices to show (1.3). Observe that

$$\frac{\frac{1}{g(t)} - \frac{1}{g(x)}}{t - x} = \frac{g(x) - g(t)}{g(t)g(x)(t - x)}.$$

Thus by Theorem 1.12 and Theorem 1.21 again, we have

$$\begin{aligned} \lim_{t \rightarrow x} \frac{\frac{1}{g(t)} - \frac{1}{g(x)}}{t - x} &= \lim_{t \rightarrow x} \frac{g(x) - g(t)}{g(t)g(x)(t - x)}, \\ &= \frac{1}{g(x)} \lim_{t \rightarrow x} \frac{1}{g(t)} \lim_{t \rightarrow x} \frac{g(x) - g(t)}{(t - x)} \\ &= -\frac{g'(x)}{g^2(x)}. \end{aligned}$$

□

Definition 1.26. For $n \in \mathbb{N}$, $a_0, a_1, \dots, a_n \in \mathbb{R}$, define

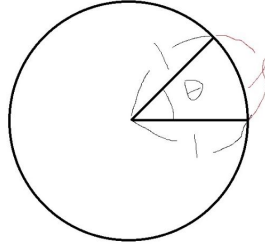
$$p(x) = a_0 + a_1x + \dots + a_nx^n \quad (x \in \mathbb{R}),$$

where $\mathbb{N}$ denotes the natural number system. Then $p(x)$ becomes a real-valued function on $\mathbb{R}$. $p(x)$ is called a **polynomial**.

Corollary 1.27. All polynomials are differentiable on $\mathbb{R}$.

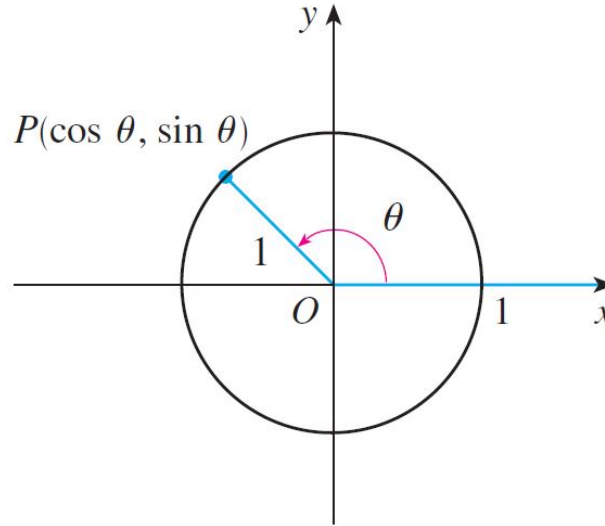
Proof. Obviously, all constant functions and the identity function x are differentiable on $\mathbb{R}$. Therefore this corollary is an easy consequence of Theorem 1.25. □

Definition 1.28. • (Angle) For two segments whose length is 1, we define an **angle** as the length of corresponding arc (see the figure below).



In other words, we define the angle θ as the length of ℓ .

- *(Cosine and Sine)* Consider a point P at the unit circle and the line segment $\overline{OP}$ from the origin to the point P . Let θ be the angle between x -axis and the line segment $\overline{OP}$. We define $\cos \theta$ and $\sin \theta$ as the x -coordinate and y -coordinate of the point P , respectively (see the following figure).



Exercise 1.29 (Difficult(?)). *Prove that*

(i)

$$\frac{d}{dx}(\cos x) = -\sin x$$

(ii)

$$\frac{d}{dx}(\sin x) = \cos x.$$

Theorem 1.30 (Chain Rule). *Let f be a real-valued function on $[a, b]$ and g be a real-valued function defined on a set I containing the range of f . Assume that f is differentiable at $x \in [a, b]$ and g is differentiable at $f(x)$. Define*

$$h(t) := g \circ f(t) := g(f(t)) \quad (a \leq t \leq b).$$

Then h is differentiable at x and

$$h'(x) = g'(f(x))f'(x),$$

i. e.

$$\frac{dh}{dt}(x) = \frac{d(g \circ f)}{dt}(x) = \frac{dg}{dt}(f(x)) \cdot \frac{df}{dt}(x).$$

Proof. (Intuitive proof)

$$\begin{aligned}
\lim_{t \rightarrow x} \frac{h(t) - h(x)}{t - x} &= \lim_{t \rightarrow x} \frac{g(f(t)) - g(f(x))}{t - x} \\
&= \lim_{t \rightarrow x} \left(\frac{g(f(t)) - g(f(x))}{f(t) - f(x)} \frac{f(t) - f(x)}{t - x} \right) \\
&= \lim_{t \rightarrow x} \left(\frac{g(f(t)) - g(f(x))}{f(t) - f(x)} \right) \left(\lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} \right) \\
&= \lim_{f(t) \rightarrow f(x)} \left(\frac{g(f(t)) - g(f(x))}{f(t) - f(x)} \right) \left(\lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} \right) \\
&= g'(f(x)) f'(x).
\end{aligned}$$

(Rigorous proof). Due to the similarity, we only consider the case $x \in (a, b)$. Recalling I is the domain of the function g , for $s \in I$ such that $s \neq f(x)$, we define

$$E_{g'}(s) := \frac{g(s) - g(f(x))}{s - f(x)} - g'(f(x))$$

and for $s = f(x)$

$$E_{g'}(s) := 0.$$

Due to the similarity, we only consider the case $f(x) \in (c, d) \subset I$ for some $c < d$. Then

$$g(s) - g(f(x)) = (s - f(x))E_{g'}(s) + (s - f(x))g'(f(x)) \quad \forall s \in I$$

and $E_{g'}(s) \rightarrow 0$ as $s \rightarrow f(x)$. Moreover, f is continuous at x by Theorem 1.21, we have

$$\lim_{t \rightarrow x} E_{g'}(f(t)) = \lim_{s \rightarrow f(x)} E_{g'}(s) = 0. \quad (1.5)$$

Therefore

$$\begin{aligned}
\lim_{t \rightarrow x} \frac{h(t) - h(x)}{t - x} &= \lim_{t \rightarrow x} \frac{g(f(t)) - g(f(x))}{t - x} \\
&= \lim_{t \rightarrow x} \frac{(f(t) - f(x))E_{g'}(f(t)) + (f(t) - f(x))g'(f(x))}{t - x} \\
&= \lim_{t \rightarrow x} \frac{(f(t) - f(x))}{t - x} \lim_{t \rightarrow x} E_{g'}(f(t)) + \lim_{t \rightarrow x} \frac{(f(t) - f(x))}{t - x} \lim_{t \rightarrow x} g'(f(x)) \\
&= f'(x) \cdot 0 + f'(x)g'(f(x)) = f'(x)g'(f(x)).
\end{aligned}$$

□

Exercise 1.31. Prove (1.5).

Exercise 1.32. Explain why the intuitive proof in Theorem 1.30 is not rigorous.

Exercise 1.33 (Difficult(?)). Let $p \in \mathbb{Q}$ and define

$$f(x) = x^p,$$

where $\mathbb{Q}$ denotes the rational number system. Then for all $x > 0$, f is differentiable at x and

$$f'(x) = px^{p-1}.$$

Example 1.34. (i) Define

$$f(x) = |x|.$$

Then

$$f'(x) = \begin{cases} 1 & (x > 0) \\ -1 & (x < 0). \end{cases}$$

and f is not differentiable at $x = 0$.

(ii) Let $p \in (1, \infty)$ and define

$$f(x) = |x|^p.$$

Then

$$f'(x) = \begin{cases} px^{p-1} & (x > 0) \\ -px^{p-1} & (x < 0). \end{cases}$$

Moreover, f is differentiable at $x = 0$ and $f'(x) = 0$. Indeed,

$$\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t - 0} = \lim_{t \rightarrow 0} |t|^{p-1} = 0.$$

(iii) Define

$$f(x) = \begin{cases} x \sin \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0). \end{cases}$$

For $x \neq 0$, by Theorem 1.25 and Theorem 1.30,

$$f'(x) = \sin \frac{1}{x} + x \cos \frac{1}{x} \left(-\frac{1}{x^2} \right) = \sin \frac{1}{x} - \frac{1}{x} \cos \frac{1}{x}.$$

For $t \neq 0$, observe that

$$\frac{f(t) - f(0)}{t - 0} = \frac{t \sin \frac{1}{t}}{t} = \sin \frac{1}{t}.$$

Therefore

$$\lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t - 0}$$

does not exist and thus f is not differentiable at $x = 0$.

(iv) Define

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0). \end{cases}$$

For $x \neq 0$, by Theorem 1.25 and Theorem 1.30,

$$f'(x) = 2x \sin \frac{1}{x} + x^2 \cos \frac{1}{x} \left(-\frac{1}{x^2} \right) = 2x \sin \frac{1}{x} - \cos \frac{1}{x}.$$

For $t \neq 0$, observe that

$$\frac{f(t) - f(0)}{t - 0} = \frac{t^2 \sin \frac{1}{t}}{t} = t \sin \frac{1}{t}.$$

Thus

$$f'(0) = \lim_{t \rightarrow 0} \frac{f(t) - f(0)}{t - 0} = 0.$$

Therefore, we have

$$f'(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0). \end{cases}$$

However, $\lim_{x \rightarrow 0} f'(x)$ does not exist and thus f' is not continuous at 0.

Definition 1.35. Let f be a real-valued function defined on a metric space X with the metric d . We say that f has a **local maximum** at a point $p \in X$ iff there exists $\delta > 0$ such that $f(q) \leq f(p)$ for all $q \in X$ with $d(p, q) < \delta$.

Similarly, we can define a local minimum.

Exercise 1.36. Define a local minimum.

Theorem 1.37. Let f be a real-valued function defined on $[a, b]$. Assume that f has a local maximum at a point $x \in (a, b)$ and f is differentiable at x . Then

$$f'(x) = 0.$$

Proof. By Definition 1.35, there exists a $\delta_0 > 0$ such that

$$f(t) \leq f(x)$$

for all $t \in (x - \delta_0, x + \delta_0) \cap [a, b]$. Since $x \in (a, b)$ and (a, b) is an open set, there exists an $\delta_1 > 0$ such that

$$(x - \delta_1, x + \delta_1) \subset (a, b).$$

Set $\delta := \min\{\delta_0, \delta_1\}$. Then

$$a < x - \delta < x < x + \delta < b$$

and

$$f(t) \leq f(x)$$

for all $t \in (x - \delta, x + \delta)$. Thus for all $t \in (x - \delta, x)$, we have

$$\frac{f(t) - f(x)}{t - x} \geq 0$$

and on the other hand for all $t \in (x, x + \delta)$

$$\frac{f(t) - f(x)}{t - x} \leq 0.$$

Since $f'(x)$ exists and

$$\lim_{t \rightarrow x^+} \frac{f(t) - f(x)}{t - x} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} = \lim_{t \rightarrow x^-} \frac{f(t) - f(x)}{t - x},$$

we finally have

$$0 \geq \lim_{t \rightarrow x^+} \frac{f(t) - f(x)}{t - x} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x} = \lim_{t \rightarrow x^-} \frac{f(t) - f(x)}{t - x} \geq 0$$

and it clearly implies that $f'(x) = 0$. □

Corollary 1.38. Let f be a real-valued function defined on $[a, b]$. Assume that f has a local minimum at a point $x \in (a, b)$ and f is differentiable at x . Then

$$f'(x) = 0.$$

Proof. Define $g(x) = -f(x)$. Then obviously g has a local maximum at the point x . Therefore by Theorem 1.37, we have

$$-f'(x) = g'(x) = 0.$$

□

Exercise 1.39. *Prove or disprove that if f has a local minimum at a point $x \in (a, b)$, then*

$$f'(x) = 0.$$

Theorem 1.40 (Rolle's theorem). *Let f be a real-valued function on $[a, b]$. Assume that f is continuous on $[a, b]$, differentiable on (a, b) , and*

$$f(a) = f(b).$$

Then there exists a $x \in (a, b)$ such that

$$f'(x) = 0.$$

Proof. One of the following three cases must be satisfied:

- (i) there exists a $t \in (a, b)$ such that $f(t) > f(a) = f(b)$
- (ii) there exists a $t \in (a, b)$ such that $f(t) < f(a) = f(b)$
- (iii) $f(t) = f(a) = f(b)$ for all $t \in (a, b)$.

We prove the theorem depending on a case. Before going further, note that f has a maximum at $x_1 \in [a, b]$ and a minimum at $x_2 \in [a, b]$ since f is continuous on $[a, b]$ and $[a, b]$ is a compact set.

- (Case (i)) Since there exists a $t \in (a, b)$ such that $f(t) > f(a) = f(b)$, we have $x_1 \in (a, b)$. Thus by Theorem 1.37,

$$f'(x_1) = 0.$$

- (Case(ii)) Since there exists a $t \in (a, b)$ such that $f(t) < f(a) = f(b)$, we have $x_2 \in (a, b)$. Thus by Theorem 1.37 again,

$$f'(x_2) = 0$$

- (Case(iii)) Since f is a constant function, $f'(x) = 0$ for all $x \in [a, b]$.

□

Theorem 1.41 (Mean-value theorem). *Let f be a real-valued function on $[a, b]$. Assume that f is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a $x \in (a, b)$ such that*

$$\frac{f(b) - f(a)}{b - a} = f'(x).$$

Proof. Define

$$h(t) = \frac{f(b) - f(a)}{b - a}(t - a) + f(a) - f(t) \quad (a \leq t \leq b).$$

Then obviously h is continuous on $[a, b]$ and differentiable on (a, b) . Moreover,

$$h(a) = h(b).$$

Therefore by Rolle's theorem, there exists a $x \in (a, b)$ such that

$$h'(x) = 0,$$

which implies

$$\frac{f(b) - f(a)}{b - a} = f'(x).$$

The theorem is proved. $\square$

Theorem 1.42 (Cauchy's mean-value theorem). *Let f and g be real-valued functions on $[a, b]$. Assume that both f and g are continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a $x \in (a, b)$ such that*

$$[f(b) - f(a)]g'(x) = [g(b) - g(a)]f'(x).$$

Proof. Define

$$h(t) = [f(b) - f(a)](g(t) - g(a)) - [g(b) - g(a)](f(t) - f(a)) \quad (a \leq t \leq b).$$

Then obviously h is continuous on $[a, b]$ and differentiable on (a, b) . Moreover, one can easily check that

$$h(a) = h(b).$$

Therefore by Rolle's theorem, there exists a $x \in (a, b)$ such that

$$h'(x) = 0,$$

which implies

$$[f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x) = 0.$$

$\square$

Definition 1.43. *Let f be a real-valued function on a subset I of $\mathbb{R}$. We say that*

(i) f is **monotonically increasing** on I iff

$$f(x) \leq f(y)$$

whenever $x, y \in I$ and $x < y$.

(ii) f is **strictly increasing** on I iff

$$f(x) < f(y)$$

whenever $x, y \in I$ and $x < y$.

(iii) f is **monotonically decreasing** on I iff

$$f(x) \geq f(y)$$

whenever $x, y \in I$ and $x < y$.

(iv) f is **strictly decreasing** on I iff

$$f(x) > f(y)$$

whenever $x, y \in I$ and $x < y$.

Theorem 1.44. *Let f be a real-valued differentiable function on (a, b) . Then the following three statements hold:*

(i) *If $f'(x) \geq 0$ for all $x \in (a, b)$, then f is monotonically increasing on (a, b) .*

(ii) *If $f'(x) \leq 0$ for all $x \in (a, b)$, then f is monotonically decreasing on (a, b) .*

(iii) *If $f'(x) = 0$ for all $x \in (a, b)$, then f is a constant.*

Proof. Let $a < x < y < b$. Then f is differentiable on (x, y) and continuous on $[x, y]$. Thus by the mean-value theorem, there exists a $z \in (x, y)$ such that

$$f(y) - f(x) = (y - x)f'(z). \quad (1.6)$$

All assertions (i), (ii), and (iii) are easily obtained from (1.6). $\square$

Exercise 1.45. Let f be a real-valued differentiable function on (a, b) . Prove or disprove the following statements hold:

- (i) If f is monotonically increasing on (a, b) , then $f'(x) \geq 0$ for all $x \in (a, b)$.
- (ii) If f is monotonically decreasing on (a, b) , then $f'(x) \leq 0$ for all $x \in (a, b)$.

Exercise 1.46. Let f be a real-valued continuous function on (a, b) . Prove or disprove the following two statements hold:

- (i) If f is monotonically increasing on (a, b) , then $f'(x) \geq 0$ for all $x \in (a, b)$.
- (ii) If f is monotonically decreasing on (a, b) , then $f'(x) \leq 0$ for all $x \in (a, b)$.

Exercise 1.47. Let f be a real-valued differentiable function on (a, b) . Prove or disprove the following two statements hold:

- (i) If $f'(x) > 0$ for all $x \in (a, b)$, then f is strictly increasing on (a, b) .
- (ii) If $f'(x) < 0$ for all $x \in (a, b)$, then f is strictly decreasing on (a, b) .

Recall the definition of the supremum and the infimum.

Definition 1.48. Let $A \subset \mathbb{R}$.

- A real number $u \in \mathbb{R}$ is called **upper bound** of A iff

$$a \leq u \quad \forall a \in A.$$

- A real number $\ell \in \mathbb{R}$ is called **lower bound** of A iff

$$\ell \leq a \quad \forall a \in A.$$

- A set A is called **bounded above** iff A has an upper bound, i.e. there exists an upper bound of A .
- A set A is called **bounded below** iff A has a lower bound, i.e. there exists a lower bound of A .
- An upper bound $\underline{u}$ of A is called **the least upper bound** or **the supremum** of A iff

$$\underline{u} \leq u$$

for all upper bounds u of A . We write

$$\sup A := \underline{u}.$$

- A lower bound $\bar{\ell}$ of A is called **the greatest lower bound** or **the infimum** of A iff

$$\ell \leq \bar{\ell}$$

for all lower bounds ℓ of A . We write

$$\inf A := \bar{\ell}.$$

Remark 1.49. • Any subset $A \subset \mathbb{R}$ which is nonempty and bounded above has a supremum in $\mathbb{R}$, i.e. $\sup A$ exists in $\mathbb{R}$. Moreover, for any $\varepsilon > 0$, there exists a $y \in A$ such that

$$\sup A - \varepsilon < y.$$

- Any subset $A \subset \mathbb{R}$ which is nonempty and bounded below has a infimum, i.e. $\inf A$ exists. Moreover, for any $\varepsilon > 0$, there exists a $y \in A$ such that

$$y < \inf A + \varepsilon.$$

Lemma 1.50. *Let f be a real-valued continuous function on $[a, b]$ with $a < b$. Assume that there is a real number λ such that*

$$f(a) < \lambda < f(b).$$

Define

$$A := \{t \in [a, b] : f(t) \leq \lambda\}.$$

Then $\sup A \in (a, b)$.

Proof. Obviously, A is nonempty and bounded above since $a \in A$ and $A \subset [a, b]$. Thus $\sup A$ exists in $\mathbb{R}$. Put $x = \sup A$. Then $x \in [a, b]$ since x is an upper bound of A and b is an upper bound of A . Observe that for all $y \in (x, b]$,

$$f(y) > \lambda. \quad (1.7)$$

Let $\varepsilon > 0$ be given. Since f is continuous at x , there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon \quad (1.8)$$

for all $y \in (x - \delta, x + \delta) \cap [a, b]$. To complete the proof, we use the reduction to absurdity. First assume that $x = a$. Take $\varepsilon = \frac{\lambda - f(a)}{2}$ and choose a $\delta > 0$ and a $y \in (a, a + \delta) \cap [a, b]$ so that (1.8) holds. Thus by (1.7) and (1.8)

$$\frac{\lambda}{2} + \frac{f(a)}{2} = \lambda - \frac{\lambda - f(a)}{2} < f(y) - \frac{\lambda - f(a)}{2} = f(y) - \varepsilon < f(x) = f(a).$$

Therefore, we have $\lambda < f(a)$ and it is contradiction. Next we assume $x = b$. Take $\varepsilon = \frac{f(b) - \lambda}{2}$ and choose a $\delta > 0$ so that (1.8) holds. Since $x = \sup A$, there exists a $y \in A$ such that

$$x - \delta < y.$$

Thus by (1.8) and the definition of A ,

$$f(b) = f(x) < f(y) + \varepsilon \leq \lambda + \varepsilon = \lambda + \frac{f(b) - \lambda}{2} = \frac{f(b) + \lambda}{2}.$$

Therefore, we have

$$f(b) < \lambda,$$

which is contradiction. The lemma is proved. $\square$

Theorem 1.51 (Intermediate value theorem). *Let f be a real-valued continuous function on $[a, b]$ with $a < b$. Assume that there is a real number λ such that*

$$f(a) < \lambda < f(b).$$

Then there is a $x \in (a, b)$ such that

$$f(x) = \lambda.$$

Proof. Define

$$A := \{t \in [a, b] : f(t) \leq \lambda\}.$$

and put $x = \sup A$. Then by Lemma 1.50, $x \in (a, b)$. Thus it is sufficient to show that $f(x) = \lambda$. Let ε be an arbitrary positive constant. Since f is continuous at x , there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon \quad (1.9)$$

for all $y \in (x - \delta, x + \delta) \cap [a, b]$. Moreover, since $x = \sup A$, there exists a $y^* \in A$ such that

$$x - \delta < y^* \leq x.$$

By the definition of A , we have

$$f(y^*) \leq \lambda. \quad (1.10)$$

Combining (1.9) and (1.10), we have

$$f(x) < f(y^*) + \varepsilon \leq \lambda + \varepsilon. \quad (1.11)$$

On the other hand, since $x \in (a, b)$ there exists a $y^{**} \in (x, x + \delta) \cap [a, b]$. Since $y^{**} > x = \sup A$, we have

$$f(y^{**}) > \lambda \quad (1.12)$$

Combining (1.9) and (1.12), we obtain

$$\lambda - \varepsilon < f(y^{**}) - \varepsilon < f(x). \quad (1.13)$$

Finally (1.11) and (1.13) imply

$$\lambda - \varepsilon < f(x) < \lambda + \varepsilon.$$

Since $\varepsilon > 0$ can be any arbitrary positive constant, we conclude that

$$f(x) = \lambda.$$

The theorem is proved. $\square$

Exercise 1.52. Let f be a real-valued continuous function on $[a, b]$ with $a < b$. Assume that there is a real number λ such that

$$f(a) < \lambda < f(b).$$

Define

$$A := \{t \in [a, b] : f(t) > \lambda\}.$$

Prove that $\inf A$ exists in $\mathbb{R}$ and

$$f(x) = \lambda.$$

Recall that the derivative of

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0). \end{cases}$$

is given by

$$f'(x) = \begin{cases} 2x \sin \frac{1}{x} - \cos \frac{1}{x} & (x \neq 0) \\ 0 & (x = 0). \end{cases}$$

Obviously $f'(x)$ is not continuous at x . However, $f'(x)$ satisfies the intermediate value property, i.e. for all $a < b$ and $\lambda \in \mathbb{R}$ such that

$$f'(a) < \lambda < f'(b),$$

there exists a $x \in (a, b)$ such that

$$f'(x) = \lambda.$$

Thus it is natural to ask if it is coincidence? To answer this question, we need the following lemma.

Lemma 1.53. *Let f be a real-valued differentiable function on $[a, b]$. Then*

- (i) *if $f'(a) < 0$ then there exists a $t_1 \in (a, b)$ such that $f(t_1) < f(a)$.*
- (ii) *if $f'(b) > 0$ then there exists a $t_2 \in (a, b)$ such that $f(t_2) < f(b)$.*

Proof. Due to the similarity, we only prove the statement (i). By the definition of the derivative

$$f'(a) = \lim_{t \rightarrow a^+} \frac{f(t) - f(a)}{t - a}$$

and the right limit, for any $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\left| \frac{f(t) - f(a)}{t - a} - f'(a) \right| < \varepsilon$$

for all $t \in (a, a + \delta)$. In particular, for $\varepsilon = -\frac{f'(a)}{2}$, there exists a $\delta > 0$ such that

$$\left| \frac{f(t) - f(a)}{t - a} - f'(a) \right| < -\frac{f'(a)}{2} \quad (1.14)$$

for all $t \in (a, a + \delta)$. Thus one can find a $t_1 \in (a, a + \delta)$ so that (1.14) holds with $t = t_1$ and (1.14) implies

$$\frac{f(t_1) - f(a)}{t_1 - a} < \frac{f'(a)}{2} < 0.$$

Finally we have $f(t_1) < f(a)$ since $t_1 > a$. □

Exercise 1.54. *Prove (ii) in Lemma 1.53 in the following two ways:*

- (1) *from the definition of the derivative*
- (2) *by applying the statement (i).*

Theorem 1.55 (Darboux Theorem). *Let f be a real-valued differentiable function on $[a, b]$. Then the derivative $f'(x)$ satisfies the intermediate value property, i.e. for any λ such that*

$$f'(a) < \lambda < f'(b),$$

there exists a $x \in (a, b)$ satisfying $f'(x) = \lambda$.

Proof. Define

$$g(t) := f(t) - \lambda t \quad (t \in [a, b]).$$

Then g is differentiable on $[a, b]$ and by Theorem 1.21, g is continuous on $[a, b]$. Thus there exists a $x \in [a, b]$ such that $g(x) = \min\{g(t) : t \in [a, b]\}$ since $[a, b]$ is a compact set. Moreover, one can easily check that $g'(a) = f'(a) - \lambda < 0$ and $g'(b) = f'(b) - \lambda > 0$. Thus by Lemma 1.53, there exist $t_1 \in (a, b)$ and $t_2 \in (a, b)$ such that $g(t_1) < g(a)$ and $g(t_2) < g(b)$. These two inequalities imply that $g(x) =$

$\min\{g(t) : t \in [a, b]\} \leq g(t_1) < g(a)$ and $g(x) = \min\{g(t) : t \in [a, b]\} \leq g(t_2) < g(b)$. Therefore $x \in (a, b)$ and g has a local minimum at $x \in (a, b)$. Finally, by Theorem 1.37, we have $g'(x) = 0$ and thus

$$f'(x) = \lambda.$$

The theorem is proved. $\square$

Exercise 1.56. Let f be a real-valued differentiable function on $[a, b]$. Prove that for any λ such that

$$f'(b) < \lambda < f'(a),$$

there exists a $x \in (a, b)$ satisfying $f'(x) = \lambda$.

Corollary 1.57. Let f be a real-valued differentiable function on $[a, b]$. Assume that f' is not right continuous at $x \in [a, b]$. Then the right limit $\lim_{t \rightarrow x^+} f'(t)$ does not exist.

Proof. If $x = b$, then the right limit of f at x cannot be defined and thus obviously the right limit does not exist. So we may assume $x \in [a, b)$. We use the reduction to absurdity, that is, assume that

$$\lim_{t \rightarrow x^+} f'(t) \text{ exists.}$$

Since f' is not right continuous,

$$f'(x) \neq \lim_{t \rightarrow x^+} f'(t).$$

We additionally assume that

$$f'(x) < \lim_{t \rightarrow x^+} f'(t)$$

without loss of generality since we can consider the function $-f$. Choose a λ such that

$$f'(x) < \lambda < \lim_{t \rightarrow x^+} f'(t).$$

Then there exists a $\delta > 0$ such that

$$f'(x) < \lambda < f'(t) \tag{1.15}$$

for all $t \in (x, x + \delta)$. However, by Theorem 1.55, there exists a $x_1 \in (x, x + \delta)$ such that

$$f'(x_1) = \lambda,$$

which leads to contradiction that $f'(x_1) < f'(x_1)$ due to (1.15). $\square$

Exercise 1.58. Let f be a real-valued function on $[a, b]$ and assume that $\lim_{t \rightarrow x^+} f(t)$ exists at $x \in [a, b)$. Then for any $\lambda < \lim_{t \rightarrow x^+} f(t)$, there exists a $\delta > 0$ such that

$$\lambda < f(t)$$

for all $t \in (x, x + \delta)$.

Exercise 1.59. Let f be a real-valued differentiable function on $[a, b]$. Assume that f' is not left continuous at $x \in [a, b]$. Prove that the left limit $\lim_{t \rightarrow x^-} f'(t)$ does not exist.

Exercise 1.60. Let f be a real-valued differentiable function on $[a, b]$. Assume that f' is not continuous at $x \in [a, b]$. Prove that the limit $\lim_{t \rightarrow x} f'(t)$ does not exist.

Definition 1.61. Let f be a real-valued function defined on an open interval (a, b) with $a < b$. For $c \in (a, b)$, we write

$$\lim_{x \rightarrow c} f(x) = \infty$$

or $f(x) \rightarrow \infty$ as $x \rightarrow c$ iff for any $M > 0$, there exists a $\delta > 0$ such that

$$f(y) > M$$

for all $y \in (c - \delta, c) \cup (c, c + \delta)$. On the other hand, we write

$$\lim_{x \rightarrow c} f(x) = -\infty$$

or $f(x) \rightarrow -\infty$ as $x \rightarrow c$ iff for any $M < 0$, there exists a $\delta > 0$ such that

$$f(y) < M$$

for all $y \in (c - \delta, c) \cup (c, c + \delta)$. Similarly, for $c \in [a, b)$

$$\lim_{x \rightarrow c^+} f(x) = \infty$$

or $f(x) \rightarrow \infty$ as $x \downarrow c$ iff for any $M > 0$, there exists a $\delta > 0$ such that

$$f(y) > M$$

for all $y \in (c, c + \delta)$.

Exercise 1.62. Define

$$\lim_{x \rightarrow c^+} f(x) = -\infty, \quad \lim_{x \rightarrow c^-} f(x) = \infty,$$

and

$$\lim_{x \rightarrow c^-} f(x) = -\infty.$$

Exercise 1.63. Let f be a real-valued function defined on an open interval (a, b) with $a < b$ and $c \in (a, b)$. Prove

(i)

$$\lim_{x \rightarrow c^+} f(x) = \infty$$

iff for all $M \in \mathbb{R}$, there exists a $\delta > 0$ such that

$$f(y) > M$$

for all $y \in (c, c + \delta)$.

(ii)

$$\lim_{x \rightarrow c^+} f(x) = \infty$$

iff there exists $M_0 \in \mathbb{R}$ such that for all $M > M_0$, one can find a $\delta > 0$ such that

$$f(y) > M$$

for all $y \in (c, c + \delta)$.

Remark 1.64. Let f be a real-valued function defined on an open interval (a, b) , where $-\infty \leq a < b \leq \infty$. We can define all limits in a similar way even though $a = -\infty$ or $b = \infty$.

Lemma 1.65. *Let f be a real-valued function defined on an open interval (a, b) with $-\infty \leq a < b \leq \infty$ and $A \in \mathbb{R}$. Then*

$$\lim_{t \rightarrow a^+} f(t) = A$$

iff for all real numbers q_1 and q_2 such that $q_1 < A < q_2$, there exist exists a $c \in (a, b)$ such that

$$q_1 < f(y) < q_2.$$

for all $y \in (a, c)$.

Proof. We only consider the case $-\infty < a$ and leave the case $a = -\infty$ as an exercise. First we prove “the if part ($\Leftarrow$)”. Let $\varepsilon > 0$ be given. Choose q_1 and q_2 so that

$$q_1 \in (A - \varepsilon, A) \tag{1.16}$$

and

$$q_2 \in (A, A + \varepsilon). \tag{1.17}$$

Then by the assumption, there exists a $c \in (a, b)$ such that

$$q_1 < f(y) < q_2$$

for all $y \in (a, c)$. Thus

$$f(y) - A < q_2 - A \tag{1.18}$$

and

$$A - f(y) < A - q_1 \tag{1.19}$$

for all $y \in (a, c)$. Applying (1.18), (1.19), (1.16), and (1.17), we have

$$|f(y) - A| < \max\{q_2 - A, A - q_1\} < \varepsilon$$

for all $y \in (a, c)$. Finally taking $\delta = c - a$, we have

$$|f(y) - A| < \varepsilon$$

for all $y \in (a, a + \delta)$. Next we prove “the only if part ($\Rightarrow$)”. Let $q_1 < A < q_2$. Define

$$\varepsilon = \min\{A - q_1, q_2 - A\}.$$

Then obviously $\varepsilon > 0$ and for this ε , there exists a $\delta > 0$ such that

$$|f(y) - A| < \varepsilon$$

for all $y \in (a, a + \delta)$. Choose a $c \in (a, a + \delta) \cap (a, b)$. Then

$$|f(y) - A| < \varepsilon$$

for all $y \in (a, c)$. This inequality implies

$$f(y) = f(y) - A + A \leq |f(y) - A| + A < \varepsilon + A \leq q_2 - A + A = q_2$$

and

$$f(y) = f(y) - A + A > -\varepsilon + A \geq -(A - q_1) + A = q_1$$

for all $y \in (a, c)$. The lemma is proved. $\square$

Exercise 1.66. Let f be a real-valued function on an open interval (a, b) . Suppose that

$$\lim_{t \rightarrow a^+} f(t) = \infty.$$

Show that for any $x \in (a, b)$, there exists $c \in (a, b)$ such that

$$f(y) > f(x) \quad \forall y \in (a, c).$$

Exercise 1.67. Prove lemma 1.65 when $a = -\infty$.

Theorem 1.68 (L'Hospital's rule). Let f and g be real-valued differentiable functions on (a, b) , where

$$-\infty \leq a < b \leq \infty.$$

Assume that $g'(x) \neq 0$ for all $x \in (a, b)$ and

$$\lim_{t \rightarrow a^+} \frac{f'(t)}{g'(t)} = A, \quad (1.20)$$

for some $A \in [-\infty, \infty] := \mathbb{R} \cup \{-\infty, \infty\}$. Additionally, suppose that

$$\lim_{t \rightarrow a^+} f(t) = \lim_{t \rightarrow a^+} g(t) = 0 \quad (1.21)$$

or

$$\lim_{t \rightarrow a^+} g(t) = \infty. \quad (1.22)$$

Then

$$\lim_{t \rightarrow a^+} \frac{f(t)}{g(t)} = A.$$

Proof. We divide the proof into several steps.

- (i) Assume $-\infty < A < \infty$. In this step, we prove for any real number q_2 such that $A < q_2$, there exists a $c \in (a, b)$ such that

$$\frac{f(y)}{g(y)} < q_2 \quad \forall y \in (a, c).$$

Let $A < q_2$ and choose $r \in (A, q_2)$. By (1.20), there exists a $c_0 \in (a, b)$ such that

$$\frac{f'(t)}{g'(t)} < r \quad \forall t \in (a, c_0).$$

Let $a < x < y < c_0$. Then by Cauchy's mean-value theorem, there exists a $t_0 \in (a, c_0)$ such that

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t_0)}{g'(t_0)} < r. \quad (1.23)$$

Suppose (1.21) holds. Then taking $x \downarrow a$ in (1.23), we have

$$\frac{f(y)}{g(y)} \leq r < q_2 \quad \forall y \in (a, c_0).$$

Next suppose (1.22) holds. Fix y in (1.23) and take x close enough to a so that

$$g(x) > 0 \quad \text{and} \quad g(x) > g(y), \quad (1.24)$$

that is, there exists $c_1 \in (a, y)$ such that (1.24) holds for all $x \in (a, c_1)$. For these x , multiplying (1.23) by $\frac{g(x)-g(y)}{g(x)}$, we have

$$\frac{f(x) - f(y)}{g(x)} < r \frac{g(x) - g(y)}{g(x)}.$$

and thus

$$\frac{f(x)}{g(x)} < r - r \frac{g(y)}{g(x)} + \frac{f(y)}{g(x)}.$$

Since (1.22) holds, if x is close enough to a , then

$$-r \frac{g(y)}{g(x)} + \frac{f(y)}{g(x)} < q_2 - r,$$

that is, there exist $c_2 \in (a, c_1) \subset (a, b)$ such that

$$\frac{f(x)}{g(x)} < r - r \frac{g(y)}{g(x)} + \frac{f(y)}{g(x)} < q_2 \quad (1.25)$$

for all $x \in (a, c_2)$.

- (ii) Assume $-\infty < A \leq \infty$. In this step, we prove for any real number q_1 such that $q_1 < A$, there exists a $c \in (a, b)$ such that

$$q_1 < \frac{f(y)}{g(y)} \quad \forall y \in (a, c).$$

One can easily prove this step following the proof of (i) and thus we skip the details.

- (iii) We complete the proof in this step. If $A = -\infty$, then by (i)

$$\lim_{t \rightarrow a^+} \frac{f(t)}{g(t)} = -\infty = A.$$

If $A = \infty$, then by (ii)

$$\lim_{t \rightarrow a^+} \frac{f(t)}{g(t)} = \infty = A.$$

If $A \in \mathbb{R}$, then by (i), (ii), and Lemma 1.65,

$$\lim_{t \rightarrow a^+} \frac{f(t)}{g(t)} = A.$$

□

Exercise 1.69. Completely prove (ii) in the proof of Theorem 1.68.

Definition 1.70. Assume that f is a real-valued differentiable function on $[a, b]$ and its derivative f' is also differentiable on $[a, b]$. Then we denote the derivative of f' by f'' or $f^{(2)}$. If f'' is differentiable again, then the derivative of $f^{(2)}$ is denoted by $f^{(3)}$. Inductively, for all $n \geq 3$, $f^{(n)}$ denotes the derivative of $f^{(n-1)}$ if $f^{(n-1)}$ is differentiable. $f^{(n)}$ is called n -th derivative of f and we say that f is n -times differentiable at t if $f^{(n)}(t)$ exists at t .

Theorem 1.71 (A generalization of the mean-value theorem). *Let f be a real-valued function on $[a, b]$. Assume that for some $n \in \mathbb{N}$, $f^{(n-1)}$ is continuous on $[a, b]$ and $f^{(n)}(t)$ exists for every $t \in (a, b)$, where $f^{(0)} := f$ and $f^{(1)} := f'$. Then for all distinct $\alpha, \beta \in [a, b]$, there exist points $x_1, x_2, \dots, x_n$ between α and β such that*

$$\begin{aligned} f(\beta) - f(\alpha) &= f'(\alpha)(\beta - \alpha) + f''(\alpha)(x_1 - \alpha)(\beta - \alpha) + f^{(3)}(\alpha)(x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha) \\ &\quad + \dots + f^{(n-1)}(\alpha)(x_{n-2} - \alpha) \dots (x_3 - \alpha)(x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha) \\ &\quad + f^{(n)}(x_n)(x_{n-1} - \alpha) \dots (x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha). \end{aligned}$$

In particular, if $n = 1$, then there exists a x between α and β such that

$$f(\beta) - f(\alpha) = f'(x)(\beta - \alpha).$$

Proof. First we show that the theorem holds for $n = 1$. Let $\alpha, \beta \in [a, b]$ such that $\alpha \neq \beta$. If $n = 1$ and $\alpha < \beta$, then by the mean-value theorem, there exists a $x_1 \in (\alpha, \beta)$ such that

$$f(\beta) - f(\alpha) = f'(x_1)(\beta - \alpha).$$

If $n = 1$ and $\beta < \alpha$, then by the mean-value theorem again, there exists a $x_1 \in (\beta, \alpha)$ such that

$$f(\alpha) - f(\beta) = f'(x_1)(\alpha - \beta).$$

Moreover, since

$$\begin{aligned} f(\beta) - f(\alpha) &= -(f(\alpha) - f(\beta)) \\ &= -f'(x_1)(\alpha - \beta) = f'(x_1)(\beta - \alpha), \end{aligned}$$

we can easily check that the mean-value theorem implies that there exists a x_1 between α and β such that

$$f(\beta) - f(\alpha) = f'(x_1)(\beta - \alpha) \tag{1.26}$$

is true whether $\alpha < \beta$ or $\alpha > \beta$. Next if $n = 2$, we apply the mean-value theorem twice. By the mean-value theorem, there exists x_2 between α and x_1 such that

$$f'(x_1) = f'(x_1) - f'(\alpha) + f'(\alpha) = f''(x_2)(x_1 - \alpha) + f'(\alpha). \tag{1.27}$$

Plugging (1.27) in (1.26), we have

$$\begin{aligned} f(\beta) - f(\alpha) &= f'(x_1)(\beta - \alpha) = (f''(x_2)(x_1 - \alpha) + f'(\alpha))(\beta - \alpha) \\ &= f'(\alpha)(\beta - \alpha) + f''(x_2)(x_1 - \alpha)(\beta - \alpha). \end{aligned} \tag{1.28}$$

If $n = 3$, then we apply the mean-value theorem 3-times. By the mean-value theorem, there exists a x_3 between α and x_2 such that

$$f''(x_2) = f''(x_2) - f''(\alpha) + f''(\alpha) = f^{(3)}(x_3)(x_2 - \alpha) + f''(\alpha). \tag{1.29}$$

Plugging (1.29) in (1.28), we obtain

$$\begin{aligned} f(\beta) - f(\alpha) &= f'(\alpha)(\beta - \alpha) + f''(\alpha)(x_1 - \alpha)(\beta - \alpha) + f^{(3)}(x_3)(x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha). \end{aligned}$$

In general, for any $n \in \mathbb{N}$, applying the mean-value theorem n -times, we have

$$\begin{aligned} f(\beta) - f(\alpha) &= f'(\alpha)(\beta - \alpha) + f''(\alpha)(x_1 - \alpha)(\beta - \alpha) + f^{(3)}(\alpha)(x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha) \\ &\quad + \cdots + f^{(n-1)}(\alpha)(x_{n-1} - \alpha) \cdots (x_3 - \alpha)(x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha) \\ &\quad + f^{(n)}(x_n)(x_{n-1} - \alpha) \cdots (x_2 - \alpha)(x_1 - \alpha)(\beta - \alpha). \end{aligned}$$

The theorem is proved. $\square$

Corollary 1.72. *Let f be a real-valued differentiable function on $[a, b]$. Assume that f' is continuous on $[a, b]$ and $f''(t)$ exists for every $t \in (a, b)$. Then for all $t \in (a, b]$, there exists a $x_t \in (a, t)$ such that*

$$|f(t) - f(a) - f'(a)(t - a)| \leq |f''(x_t)|(t - a)^2.$$

Proof. By Theorem 1.71, there exist $x_1, x_2 \in (a, t)$ such that

$$f(t) - f(a) = f'(a)(t - a) + f''(x_2)(x_1 - a)(t - a).$$

Since $|x_1 - a| < |t - a|$, we have

$$|f(t) - f(a) - f'(a)(t - a)| = |f''(x_2)(x_1 - a)(t - a)| \leq |f''(x_2)||t - a|^2$$

$\square$

Due to Corollary 1.72, the first-order polynomial

$$f(a) + f'(a)(t - a)$$

is a good approximation of f if $f''(x)$ is bounded and t is near a , i.e. $\sup_{x \in (a, t)} |f''(x)| < \infty$. More generally, it is natural to ask if the polynomial

$$p(t) := \sum_{k=0}^{n-1} f^{(k)}(a)(t - a)^k$$

is the best polynomial approximation of $f(t)$ near $t = a$ if f is n -times differentiable? Here what does the “best approximation” mean? Observe that $f(a) = p(a)$. However for all $j \in \{1, 2, \dots, n-1\}$

$$p^{(j)}(a) = j!f^{(j)}(a).$$

Thus $p^{(j)}(a) \neq f^{(j)}(a)$ if $f^{(j)}(a) \neq 0$ nor $j = 1$. Next let's consider the polynomial

$$\tilde{p}(t) = \sum_{k=0}^{n-1} \frac{f^{(k)}(a)}{k!} (t - a)^k.$$

Then

$$\tilde{p}^{(j)}(a) = f^{(j)}(a) \quad \forall j \in \{0, 1, 2, \dots, n-1\}.$$

In this sense, we say that $\tilde{p}(t)$ is a better polynomial approximation than $p(t)$ near $t = a$.

Theorem 1.73 (Taylor's theorem). *Let f be a real-valued differentiable function on $[a, b]$. Assume that for some $n \in \mathbb{N}$, f^{n-1} is continuous on $[a, b]$ and $f^{(n)}(t)$ exists for every $t \in (a, b)$, where $f^{(0)} := f$ and $f^{(1)} := f'$. Then for all distinct $\alpha, \beta \in [a, b]$, there exists a point x between α and β such that*

$$f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (\beta - \alpha)^k + \frac{f^{(n)}(x)}{n!} (\beta - \alpha)^n. \quad (1.30)$$

Proof. Let $\alpha, \beta \in [a, b]$ such that $\alpha \neq \beta$. Define

$$p(t) := \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} (t - \alpha)^k \quad (t \in [a, b])$$

and

$$M := \frac{f(\beta) - p(\beta)}{(\beta - \alpha)^n}. \quad (1.31)$$

Define

$$g(t) := f(t) - p(t) - M(t - \alpha)^n \quad (a \leq t \leq b).$$

Then obviously,

$$g(\beta) = f(\beta) - p(\beta) - M(\beta - \alpha)^n = 0.$$

Moreover, observe that for $k = 0, 1, \dots, n-1$,

$$f^{(k)}(\alpha) = p^{(k)}(\alpha)$$

and thus

$$g^{(k)}(\alpha) = f^{(k)}(\alpha) - p^{(k)}(\alpha) - Mn(n-1) \cdots (n-k+1)(\alpha - \alpha)^{n-k} = 0.$$

Therefore, by Rolle's theorem, there exists a x_1 between α and β such that

$$g'(x_1) = 0.$$

Applying Rolle's theorem again, we can find a x_2 between α and x_1 such that

$$g^{(2)}(x_2) = 0.$$

Continuing in this way, we conclude that there exists a x_n between α and x_{n-1} such that

$$g^{(n)}(x_n) = 0. \quad (1.32)$$

Since

$$g^{(n)}(t) = f^{(n)}(t) - p^{(n)}(t) - Mn(n-1) \cdots 2 \cdot 1 = f^{(n)}(t) - Mn! \quad \forall t \in (a, b),$$

we have

$$M = \frac{f^{(n)}(x_n)}{n!} \quad (1.33)$$

Putting (1.33) in (1.31), we finally obtain (1.30) with $x = x_n$. $\square$

2. THE RIEMANN-STIELTJES INTEGRAL

Definition 2.1. Let $[a, b]$ be a closed interval with $a < b$. A partition P of $[a, b]$ is a finite set of points $x_0, x_1, \dots, x_n$ such that

$$a = x_0 \leq x_1 \leq \dots \leq x_{n-1} \leq x_n = b.$$

We write

$$\Delta x_i = x_i - x_{i-1} \quad (i = 1, \dots, n).$$

Let f be a real-valued bounded function on $[a, b]$ and $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. For each $i \in \{1, \dots, n\}$, we put

$$M_i := \sup_{x_{i-1} \leq x \leq x_i} f(x) := \sup\{f(x) : x_{i-1} \leq x \leq x_i\},$$

$$m_i := \inf_{x_{i-1} \leq x \leq x_i} f(x) := \inf\{f(x) : x_{i-1} \leq x \leq x_i\},$$

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i,$$

and

$$L(P, f) = \sum_{i=1}^n m_i \Delta x_i,$$

We say that $U(P, f)$ is an **upper sum** of f over P and $L(P, f)$ is a **lower sum** of f over P . Obviously,

$$L(P, f) \leq U(P, f).$$

Finally, we define

$$\int_a^{\bar{b}} f dx := \inf U(P, f)$$

and

$$\int_a^b f dx := \sup L(P, f),$$

where the $\inf$ and the $\sup$ are taken over all partitions P of $[a, b]$, i.e.

$$\inf U(P, f) := \inf\{U(P, f) : P \text{ is a partition of } [a, b]\}$$

and

$$\sup L(P, f) := \sup\{L(P, f) : P \text{ is a partition of } [a, b]\}.$$

Here $\int_a^{\bar{b}} f dx$ and $\int_a^b f dx$ are called the **upper Riemann integral** of f on $[a, b]$ and the **lower Riemann integral** of f on $[a, b]$, respectively. Moreover, obviously for any partition P of $[a, b]$,

$$L(P, f) \leq \int_a^b f dx,$$

and

$$\int_a^{\bar{b}} f dx \leq U(P, f).$$

Remark 2.2. Let f be a real-valued bounded function on $[a, b]$, i.e. there exists a positive constant $M > 0$ such that

$$|f(x)| < M \quad \forall x \in [a, b].$$

Then obviously f is bounded above and bounded below. Thus for all real-valued bounded function f and a partition P of $[a, b]$, $U(P, f)$ and $L(P, f)$ are well-defined, i.e. $U(P, f)$ and $L(P, f)$ exist in $\mathbb{R}$. More precisely,

$$\begin{aligned} -M(b-a) &= -M \sum_{i=1}^n \Delta x_i \leq \sum_{i=1}^n m_i \Delta x_i = L(P, f) \\ &\leq \sum_{i=1}^n M_i \Delta x_i = U(P, f) \leq M \sum_{i=1}^n \Delta x_i = M(b-a) \end{aligned}$$

Therefore $\int_a^b f dx$ and $\int_a^b f dx$ exist in $\mathbb{R}$ as well.

Remark 2.3. Let $P_0 = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$ and $P_1 = \{y_0, \dots, y_m\}$ be an another partition of $[a, b]$. Then $a = x_0 \leq x_1 \leq \dots \leq x_n = b$ and $a = y_0 \leq y_1 \leq \dots \leq y_m = b$. Consider the set

$$P_0 \cup P_1 = \{x_0, x_1, \dots, x_n, y_0, \dots, y_m\}.$$

Since $P_0 \cup P_1$ is a finite set, we can compare the order of all elements of $P_0 \cup P_1$. Thus we can rearrange the elements of $P_0 \cup P_1$ so that

$$P_0 \cup P_1 = \{x_0, x_1, \dots, x_n, y_0, \dots, y_m\} = \{z_0, z_1, \dots, z_{n+m+1}\}$$

and

$$a = z_0 \leq z_1 \leq z_2 \leq \dots \leq z_{n+m+1} = b.$$

Therefore we can consider $P_0 \cup P_1$ as a partition of $[a, b]$. Moreover,

$$\min\{U(P_0, f), U(P_1, f)\} \geq U(P_0 \cup P_1, f) \quad (2.1)$$

and

$$\max\{L(P_0, f), L(P_1, f)\} \leq L(P_0 \cup P_1, f). \quad (2.2)$$

Two inequalities (2.1) and (2.2) seem to be clear intuitively and a rigorous proof of these two inequalities will be given later.

Theorem 2.4. Let f be a real-valued function on $[a, b]$. Assume that $\int_a^b f dx$ and $\int_a^b f dx$ exist in $\mathbb{R}$. Then

$$\int_a^b f dx \leq \int_a^b f dx.$$

Proof. Let $\varepsilon > 0$ be given. Since

$$\int_a^b f dx := \inf U(P, f),$$

where inf is taken over all partitions of $[a, b]$, there exists a partition P_0 such that

$$U(P_0, f) < \int_a^b f dx + \varepsilon$$

Similarly, since

$$\int_a^b f dx := \sup L(P, f), \quad (2.3)$$

there exists a partition P_1 such that

$$\int_a^b f dx - \varepsilon < L(P_1, f) \quad (2.4)$$

Moreover,

$$L(P_1, f) \leq L(P_0 \cup P_1, f) \leq U(P_0 \cup P_1, f) \leq U(P_0, f). \quad (2.5)$$

Therefore, combining (2.3), (2.4), and (2.5), we have

$$\int_a^b f dx - \varepsilon < \int_a^b f dx + \varepsilon.$$

Since $\varepsilon > 0$ is an arbitrary positive constant, we finally obtain

$$\int_a^b f dx \leq \int_a^b f dx.$$

□

Corollary 2.5. *Let f be a real-valued function on $[a, b]$ and P be a partition of $[a, b]$. Assume that $\int_a^b f dx$ and $\int_a^b f dx$ exist in $\mathbb{R}$. Then*

$$L(P, f) \leq \int_a^b f dx \leq \int_a^b f dx \leq U(P, f). \quad (2.6)$$

Definition 2.6 (The Riemann integral). *Let f be a real-valued function on $[a, b]$. We say that f is Riemann-integrable on $[a, b]$ iff both $\int_a^b f dx$ and $\int_a^b f dx$ exist in $\mathbb{R}$ and*

$$\int_a^b f dx = \int_a^b f dx.$$

For any Riemann-integrable function f on $[a, b]$, we write

$$\int_a^b f dx := \int_a^b f dx.$$

Example 2.7. *Define*

$$f(x) := \begin{cases} 1 & \text{if } x \text{ is a irrational number} \\ 0 & \text{if } x \text{ is a rational number} \end{cases}$$

and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. Then

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i = \sum_{i=1}^n \Delta x_i = x_n - x_0 = b - a,$$

and

$$L(P, f) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1}^n 0 \cdot \Delta x_i = 0.$$

Therefore f is not Riemann-integrable even though f is bounded.

Example 2.8. Define

$$f(x) := \begin{cases} \frac{1}{x^2} & \text{if } x > 0 \\ 0 & \text{if } x = 0. \end{cases}$$

and let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[0, 1]$. Then there exist a $i_0 \in \{1, \dots, n\}$ such that

$$x_{i_0-1} \leq 0 < x_{i_0}.$$

Thus

$$U(P, f) = \sum_{i=1}^n M_i \Delta x_i \geq M_{i_0} \Delta x_{i_0} = \infty.$$

and it implies that $\int_a^b f dx$ does not exist in $\mathbb{R}$. However, f is Riemann-integrable on $[\delta, 1 + \delta]$ for any $\delta > 0$. Fix $\delta > 0$ to show this statement. For each $n \in \mathbb{N}$, define

$$P_n := \left\{ \delta + \frac{k}{n} : k = 0, 1, \dots, n \right\}.$$

Then P_n is a partition of $[\delta, 1 + \delta]$. Observe that for each n ,

$$U(P_n, f) = \sum_{i=1}^n M_i \Delta x_i = \sum_{i=1}^n \left(\delta + \frac{i-1}{n} \right)^{-2} \frac{1}{n} = \sum_{i=1}^n \left(\frac{n}{n\delta + i - 1} \right)^2 \frac{1}{n}$$

and

$$L(P_n, f) = \sum_{i=1}^n m_i \Delta x_i = \sum_{i=1}^n \left(\delta + \frac{i}{n} \right)^{-2} \frac{1}{n} = \sum_{i=1}^n \left(\frac{n}{n\delta + i} \right)^2 \frac{1}{n}.$$

Thus

$$\begin{aligned} U(P_n, f) - L(P_n, f) &= \frac{1}{n} \sum_{i=1}^n \left(\left(\frac{n}{n\delta + i - 1} \right)^2 - \left(\frac{n}{n\delta + i} \right)^2 \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\frac{n}{n\delta + i - 1} - \frac{n}{n\delta + i} \right) \left(\frac{n}{n\delta + i - 1} + \frac{n}{n\delta + i} \right) \\ &= \frac{1}{n} \sum_{i=1}^n \left(\frac{n}{(n\delta + i - 1)(n\delta + i)} \right) \left(\frac{n}{n\delta + i - 1} + \frac{n}{n\delta + i} \right) \\ &\leq \frac{1}{n} \sum_{i=1}^n \left(\frac{n}{(n\delta)(n\delta)} \right) \left(\frac{n}{n\delta} + \frac{n}{n\delta} \right) \\ &= \frac{2}{\delta^3} \frac{1}{n}. \end{aligned}$$

Moreover, due to (2.6), we have

$$\left| \int_{\delta}^{\delta+1} f dx - \int_{\delta}^{\delta+1} f dx \right| = \int_{\delta}^{\delta+1} f dx - \int_{\delta}^{\delta+1} f dx \leq U(P_n, f) - L(P_n, f) \leq \frac{2}{\delta^3} \frac{1}{n}$$

for all $n \in \mathbb{N}$. Finally taking $n \rightarrow \infty$ in the above inequality, we have

$$\int_{\delta}^{\delta+1} f dx = \int_{\delta}^{\delta+1} f dx$$

and thus f is Riemann-integrable on $[\delta, 1 + \delta]$.

Exercise 2.9. Define

$$f(x) := \begin{cases} \frac{1}{\sqrt{x}} & \text{if } x > 0 \\ 0 & \text{if } x = 0. \end{cases}$$

Prove or disprove that f is Riemann integrable on $[0, 1]$.

Exercise 2.10. Let f be a real-valued function on $[a, b]$. Assume that f is unbounded on $[a, b]$, i.e. there is no $M > 0$ such that

$$|f(x)| < M \quad \forall x \in [a, b].$$

Show that f is not Riemann-integrable on $[a, b]$.

Definition 2.11. Let α be a monotonically increasing function on $[a, b]$ and $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. For each $i \in \{1, \dots, n\}$, we write

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}).$$

Obviously $\Delta\alpha_i \geq 0$ for all i . Let f be a bounded function on $[a, b]$. Define

$$\begin{aligned} U(P, f, \alpha) &= \sum_{i=1}^n M_i \Delta\alpha_i, \\ L(P, f, \alpha) &= \sum_{i=1}^n m_i \Delta\alpha_i, \end{aligned} \tag{2.7}$$

$$\int_a^{\bar{b}} f d\alpha := \inf U(P, f, \alpha),$$

and

$$\int_a^b f d\alpha := \sup L(P, f, \alpha),$$

where

$$\begin{aligned} M_i &:= \sup_{x_{i-1} \leq x \leq x_i} f(x) := \sup\{f(x) : x_{i-1} \leq x \leq x_i\}, \\ m_i &:= \inf_{x_{i-1} \leq x \leq x_i} f(x) := \inf\{f(x) : x_{i-1} \leq x \leq x_i\}, \end{aligned}$$

and the $\inf$ and the $\sup$ are taken over all partitions P of $[a, b]$. We call $\int_a^{\bar{b}} f d\alpha$ and $\int_a^b f d\alpha$ the **upper and lower Riemann-Stieltjes integral** of f with respect to α on $[a, b]$, respectively. Obviously, one can easily check that for any partition P

$$L(P, f, \alpha) \leq U(P, f, \alpha). \tag{2.8}$$

Definition 2.12. We say that a function f on $[a, b]$ is **Riemann-Stieltjes integrable** with respect to α on $[a, b]$ or is **integrable with respect to α in the Riemann-Stieltjes sense** on $[a, b]$ if and only if both $\int_a^{\bar{b}} f d\alpha$ and $\int_a^b f d\alpha$ exist in $\mathbb{R}$ and

$$\int_a^{\bar{b}} f d\alpha = \int_a^b f d\alpha.$$

If f on $[a, b]$ is Riemann-Stieltjes integrable with respect to α , then we use the notation

$$\int_a^b f d\alpha$$

or

$$\int_a^b f(x) d\alpha(x) \quad (2.9)$$

to denote the value

$$\int_a^b f d\alpha = \int_a^b f d\alpha.$$

$\int_a^b f d\alpha$ is called the **Riemann-Stieltjes integral** of f with respect to α on $[a, b]$.

Remark 2.13. • Recall that both $\sum_{i=1}^3 c_i$ and $\sum_{j=1}^3 c_j$ denote the value

$$c_1 + c_2 + c_3.$$

Similarly, we can use y instead of x in (2.9). This type variable is called **dummy variable**.

- Let $\alpha(x) = x$. Then obviously, f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ if and only if f is Riemann integrable on $[a, b]$. Moreover,

$$\int_a^b f d\alpha = \int_a^b f dx.$$

Definition 2.14. Let P and P^* be partitions of $[a, b]$. We say that the partition P^* is a **refinement** of P if $P^* \supset P$. For given two partitions P_1 and P_2 of $[a, b]$, a partition P^* is called their **common refinement** if $P^* = P_1 \cup P_2$.

Theorem 2.15. Let f be a bounded function on $[a, b]$, P be a partition of $[a, b]$, and P^* be a refinement of P . Then

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \quad (2.10)$$

Proof. Since P is a partition of $[a, b]$, we can put

$$P = \{x_0, x_1, \dots, x_n\}$$

for some $n \in \mathbb{N}$.

- Case 1) P^* contains only one point more than P . Let x^* be the extra point. Then there exists a $i_0 \in \{1, \dots, n\}$ such that

$$x_{i_0-1} \leq x^* \leq x_{i_0}.$$

Thus

$$P^* = \{x_0, x_1, \dots, x_{i_0-1}, x^*, x_{i_0}, \dots, x_n\}. \quad (2.11)$$

Recall

$$m_{i_0} = \inf_{x_{i_0-1} \leq x \leq x_{i_0}} f(x)$$

and put

$$w_1 = \inf_{x_{i_0-1} \leq x \leq x^*} f(x)$$

and

$$w_2 = \inf_{x^* \leq x \leq x_{i_0}} f(x).$$

Then

$$w_1 \geq m_{i_0} \quad \text{and} \quad w_2 \geq m_{i_0}. \quad (2.12)$$

Therefore recalling (2.7), (2.12), and (2.11), we have

$$\begin{aligned}
L(P, f, \alpha) &= \sum_{i=1}^n m_i \Delta \alpha_i \\
&= \left(\sum_{i=1}^{i_0-1} m_i \Delta \alpha_i \right) + m_{i_0} \Delta \alpha_{i_0} + \left(\sum_{i=i_0+1}^n m_i \Delta \alpha_i \right) \\
&= \left(\sum_{i=1}^{i_0-1} m_i \Delta \alpha_i \right) + m_{i_0} (\alpha(x_{i_0}) - \alpha(x^*) + \alpha(x^*) - \alpha(x_{i_0-1})) + \left(\sum_{i=i_0+1}^n m_i \Delta \alpha_i \right) \\
&\leq \left(\sum_{i=1}^{i_0-1} m_i \Delta \alpha_i \right) + w_2 (\alpha(x_{i_0}) - \alpha(x^*)) + w_1 (\alpha(x^*) - \alpha(x_{i_0-1})) + \left(\sum_{i=i_0+1}^n m_i \Delta \alpha_i \right) \\
&= L(P^*, f, \alpha).
\end{aligned}$$

- Case 2) (General case). Since P^* is a refinement of P , there exists a $k \in \{0, 1, \dots\}$ such that P^* contains k -points more than P . If $k = 0$, then $P = P^*$. Thus (2.10) holds. If $k = 1$, (2.10) is proved in Case 1. So we may assume that $k \geq 2$. Put

$$\begin{aligned}
P^* - P &= \{x_1^*, \dots, x_k^*\}, \\
P_0^* &= P,
\end{aligned}$$

and

$$P_j^* := P \cup \{x_1^*, \dots, x_j^*\} \quad (j = 1, \dots, k).$$

Then

$$P = P_0^* \subset P_1^* \subset \dots \subset P_k^* = P^*$$

and P_i^* contains only one more point than P_{i-1}^* for all $i = 1, \dots, k$. Therefore by Case 1,

$$L(P_{i-1}^*, f, \alpha) \leq L(P_i^*, f, \alpha) \quad \forall i \in \{1, \dots, k\}.$$

In particular, we have

$$L(P, f, \alpha) = L(P_0^*, f, \alpha) \leq \dots \leq L(P_k^*, f, \alpha) = L(P^*, f, \alpha).$$

□

Corollary 2.16. *Let f be a bounded function on $[a, b]$, P be a partition of $[a, b]$, and P^* be a refinement of P . Then*

$$U(P^*, f, \alpha) \leq U(P, f, \alpha). \quad (2.13)$$

Exercise 2.17. *Prove Corollary 2.16.*

Corollary 2.18. *Let f be a bounded function on $[a, b]$. Then*

$$\int_a^b f d\alpha \leq \int_a^{\bar{b}} f d\alpha \quad (2.14)$$

and for any partition P of $[a, b]$,

$$L(P, f, \alpha) \leq \int_a^b f d\alpha \leq \int_a^{\bar{b}} f d\alpha \leq U(P, f, \alpha). \quad (2.15)$$

Proof. By the definition of $\int_a^b f d\alpha$ and $\int_a^b f d\alpha$, it is obvious that $L(P, f, \alpha) \leq \int_a^b f d\alpha$ and $\int_a^b f d\alpha \leq U(P, f, \alpha)$ for all partition P of $[a, b]$. Thus it is sufficient to show (2.14). Let P_1 and P_2 be partitions of $[a, b]$. Consider the common refinement P^* of P_1 and P_2 . Then by (2.10), (2.8), and (2.13),

$$L(P_1, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P_2, f, \alpha).$$

Thus we have

$$L(P_1, f, \alpha) \leq U(P_2, f, \alpha). \quad (2.16)$$

for all partitions P_1 and P_2 of $[a, b]$. Fix a partition P_2 . Then since P_1 can be any partition of $[a, b]$, (2.16) implies

$$\int_a^b f d\alpha \leq U(P_2, f, \alpha). \quad (2.17)$$

Moreover, since we fixed an arbitrary partition P_2 of $[a, b]$, (2.17) finally gives

$$\int_a^b f d\alpha \leq \int_a^b f d\alpha.$$

□

Theorem 2.19. *Let f be a bounded real-valued function on $[a, b]$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ if and only if for any $\varepsilon > 0$, there exists a partition P of $[a, b]$ such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon. \quad (2.18)$$

Proof. First we prove “the if part ($\Leftarrow$)”. Let $\varepsilon > 0$ be given. Then there exists a partition P of $[a, b]$ such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

Moreover, by (2.15),

$$0 \leq \int_a^b f d\alpha - \int_a^b f d\alpha \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, we have

$$\int_a^b f d\alpha = \int_a^b f d\alpha.$$

Next we prove “the only if part ($\Rightarrow$)”. Let $\varepsilon > 0$ be given. Recall that

$$\int_a^b f d\alpha = \inf U(P, f, \alpha)$$

and

$$\int_a^b f d\alpha = \sup L(P, f, \alpha),$$

where the sup and inf are taken over all partitions P of $[a, b]$. Since f is Riemann-Stieltjes integrable with respect to α on $[a, b]$,

$$\int_a^b f d\alpha = \int_a^b f d\alpha = \int_a^b f d\alpha.$$

Moreover, by properties of sup and inf, there exist partitions P_1 and P_2 such that

$$U(P_2, f, \alpha) < \inf_P U(P, f, \alpha) + \varepsilon/2 = \int_a^b f d\alpha + \varepsilon/2 = \int_a^b f d\alpha + \varepsilon/2 \quad (2.19)$$

and

$$\int_a^b f d\alpha - \varepsilon/2 = \int_a^b f d\alpha - \varepsilon/2 = \sup_P L(P, f, \alpha) - \varepsilon/2 < L(P_1, f, \alpha). \quad (2.20)$$

Let P be the common refinement of P_1 and P_2 . Then by (2.10), (2.13), (2.19), and (2.20),

$$U(P, f, \alpha) \leq U(P_2, f, \alpha) < \int_a^b f d\alpha + \varepsilon/2 < L(P_1, f, \alpha) + \varepsilon \leq L(P, f, \alpha) + \varepsilon,$$

which obviously implies (2.18). $\square$

Theorem 2.20. *Let f be a real-valued function on $[a, b]$.*

(i) *If there exists a partition P such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon, \quad (2.21)$$

for some $\varepsilon > 0$, then for any refinement P^ of P we have*

$$U(P^*, f, \alpha) - L(P^*, f, \alpha) < \varepsilon.$$

(ii) *Assume that there exists a partition $P = \{x_0, \dots, x_n\}$ such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon. \quad (2.22)$$

Then for all $s_i, t_i \in [x_{i-1}, x_i]$ ($i = 1, \dots, n$),

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta\alpha_i < \varepsilon.$$

(iii) *Assume that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and there exists a partition $P = \{x_0, \dots, x_n\}$ such that*

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Then for all $t_i \in [x_{i-1}, x_i]$ ($i = 1, \dots, n$),

$$\left| \sum_{i=1}^n f(t_i) \Delta\alpha_i - \int_a^b f d\alpha \right| < \varepsilon. \quad (2.23)$$

Proof. First we prove (i). Let P be a partition of $[a, b]$ satisfying (2.21) and P^* be a refinement of P . Then by (2.10) and (2.13),

$$L(P, f, \alpha) \leq L(P^*, f, \alpha) \leq U(P^*, f, \alpha) \leq U(P, f, \alpha).$$

Therefore,

$$U(P^*, f, \alpha) - L(P^*, f, \alpha) \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Next we prove (ii). Since $P = \{x_0, \dots, x_n\}$,

$$U(P, f, \alpha) = \sum_{i=1}^n M_i \Delta\alpha_i \quad \text{and} \quad L(P, f, \alpha) = \sum_{i=1}^n m_i \Delta\alpha_i,$$

where $M_i := \sup_{x_{i-1} \leq x \leq x_i} f(x)$ and $m_i := \inf_{x_{i-1} \leq x \leq x_i} f(x)$. Let $s_i, t_i \in [x_{i-1}, x_i]$ for all $i \in \{1, \dots, n\}$. Then by (2.22),

$$\sum_{i=1}^n |f(s_i) - f(t_i)| \Delta\alpha_i \leq \sum_{i=1}^n (M_i - m_i) \Delta\alpha_i = U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Finally we prove (iii). Recall that $P = \{x_1, \dots, x_n\}$ satisfies

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Let $t_i \in [x_{i-1}, x_i]$ for all $i \in \{1, \dots, n\}$. Then obviously

$$L(P, f, \alpha) \leq \sum_{i=1}^n f(t_i) \Delta\alpha_i \leq U(P, f, \alpha).$$

Moreover, since f is Riemann-Stieltjes integrable with respect to α on $[a, b]$,

$$L(P, f, \alpha) \leq \int_a^b f d\alpha = \int_a^b f d\alpha = \int_a^b f d\alpha \leq U(P, f, \alpha).$$

Therefore

$$\sum_{i=1}^n f(t_i) \Delta\alpha_i - \int_a^b f d\alpha \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon$$

and

$$\int_a^b f d\alpha - \sum_{i=1}^n f(t_i) \Delta\alpha_i \leq U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

These two inequalities obviously imply (2.23). $\square$

Let f be a real-valued function defined on a subset D of $\mathbb{R}$. Recall that f is continuous at $c \in D$ iff for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(c)| < \varepsilon$$

for all $x \in (c - \delta, c + \delta) \cap D$. Moreover, we say that f is continuous on D if f is continuous at c for all $c \in D$.

Definition 2.21. Let f be a real-valued function defined on $D \subset \mathbb{R}$. We say that f is **uniformly continuous** on D iff for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon$$

for all $x, y \in D$ satisfying $|x - y| < \delta$.

Exercise 2.22. Let f be a real-valued function defined on $D \subset \mathbb{R}$. Prove or disprove the following two statements:

- (1) If f is continuous on D , then f is uniformly continuous on D .
- (2) If f is uniformly continuous on D , then f is continuous on D .

Remark 2.23. $\bullet$ A set A is called an **index set** if for each $\alpha \in A$, there exists a corresponding set G_α . A set of sets is called a **collection** of sets or **family** of sets.

- Let $K \subset \mathbb{R}$. A collection of open sets of $\mathbb{R}$ $\{G_\alpha : \alpha \in A\}$ is called an **open cover** of K if G_α is open for all $\alpha \in A$ and

$$K \subset \bigcup_{\alpha \in A} G_\alpha.$$

If there exists a subset $A_0 \subset A$ such that

$$K \subset \bigcup_{\alpha \in A_0} G_\alpha,$$

then $\{G_\alpha : \alpha \in A_0\}$ is called a subcover of the open cover $\{G_\alpha : \alpha \in A\}$. Additionally, if A_0 is a finite set, then $\{G_\alpha : \alpha \in A_0\}$ is called a finite subcover of the open cover $\{G_\alpha : \alpha \in A\}$.

- A subset $K \subset \mathbb{R}$ is called compact if every open cover of K has a finite subcover.
- Heine-Borel Theorem says that any bounded and closed subset of $\mathbb{R}$ is compact. In particular, $[a, b]$ is compact for all $a < b$.

Theorem 2.24. *Let f be a real-valued function defined on a closed interval $[a, b]$. Then f is continuous on $[a, b]$ if and only if f is uniformly continuous on $[a, b]$.*

Proof. First we prove “the if part ($\Leftarrow$)”. Assume that f is uniformly continuous on $[a, b]$. Let $c \in [a, b]$ and it is sufficient to show that f is continuous at c . Let $\varepsilon > 0$ be given. Then since f is uniformly continuous on $[a, b]$, there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon$$

for all $x, y \in [a, b]$ satisfying $|x - y| < \delta$. In particular,

$$|f(x) - f(c)| < \varepsilon$$

for all $x \in [a, b]$ satisfying $|x - c| < \delta$, i.e. $x \in (c - \delta, c + \delta) \cap [a, b]$. Therefore f is continuous at c . Next we prove “the only if part ($\Rightarrow$)”. Assume that f is continuous at c for all $c \in [a, b]$. Let $\varepsilon > 0$ be given. Then for each $c \in [a, b]$, there exists a $\delta_c > 0$ such that

$$|f(x) - f(c)| < \frac{\varepsilon}{2} \quad (2.24)$$

for all $x \in (c - \delta_c, c + \delta_c) \cap [a, b]$. Note that the collection $\{(c - \delta_c/2, c + \delta_c/2) : c \in [a, b]\}$ is an open cover of $[a, b]$ since each $(c - \delta_c/2, c + \delta_c/2)$ is open and

$$[a, b] \subset \bigcup_{c \in [a, b]} (c - \delta_c/2, c + \delta_c/2).$$

Thus there exists a finite subcover

$$\{(c_1 - \delta_{c_1}/2, c_1 + \delta_{c_1}/2), \dots, (c_n - \delta_{c_n}/2, c_n + \delta_{c_n}/2)\}$$

such that

$$[a, b] \subset \bigcup_{i=1}^n (c_i - \delta_{c_i}/2, c_i + \delta_{c_i}/2) \quad (2.25)$$

since $[a, b]$ is compact. Set

$$\delta := \frac{1}{2} \min\{\delta_{c_1}, \dots, \delta_{c_n}\}.$$

Let $x, y \in [a, b]$ such that $|x - y| < \delta$. It is sufficient to show that

$$|f(x) - f(y)| < \varepsilon.$$

By (2.25), there exists a $i_0 \in \{1, \dots, n\}$ such that

$$x \in \left(c_{i_0} - \frac{\delta_{c_{i_0}}}{2}, c_{i_0} + \frac{\delta_{c_{i_0}}}{2}\right).$$

Then since

$$|x - y| < \delta \leq \frac{\delta_{c_{i_0}}}{2},$$

we have

$$y \in (c_{i_0} - \delta_{c_{i_0}}, c_{i_0} + \delta_{c_{i_0}}).$$

Thus by (2.24),

$$|f(x) - f(y)| \leq |f(x) - f(c_{i_0})| + |f(c_{i_0}) - f(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Exercise 2.25. Let f be a real-valued function defined on a compact set $K \subset \mathbb{R}$. Assume that f is continuous on K . Show that f is uniformly continuous on K .

Theorem 2.26. Let f be a continuous function on $[a, b]$ and α be a monotonically increasing function on $[a, b]$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Proof. Let $\varepsilon > 0$ be given. Due to Theorem 2.19, it is sufficient to find a partition P such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Choose a $\eta > 0$ so that

$$(\alpha(b) - \alpha(a))\eta < \varepsilon. \quad (2.26)$$

By Theorem 2.24, f is uniformly continuous on $[a, b]$. Thus there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \eta$$

for all $x, y \in [a, b]$ satisfying $|x - y| < \delta$. Choose a large $n \in \mathbb{N}$ so that

$$\frac{b-a}{n} < \delta.$$

and put

$$x_i = a + i \cdot \frac{b-a}{n} \quad (i = 0, \dots, n).$$

Then

$$P := \{x_0, x_1, \dots, x_n\}$$

is a partition of $[a, b]$ such that

$$\Delta x_i := x_i - x_{i-1} = \frac{b-a}{n} < \delta \quad \forall i.$$

Thus if $x, y \in [x_{i-1}, x_i]$ for some $i \in \{1, \dots, n\}$, then

$$f(x) - f(y) \leq |f(x) - f(y)| < \eta$$

and

$$\begin{aligned} M_i - m_i &= \sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{y \in [x_{i-1}, x_i]} f(y) \\ &= \sup_{x \in [x_{i-1}, x_i]} f(x) + \sup_{y \in [x_{i-1}, x_i]} (-f(y)) \leq \eta. \end{aligned}$$

Therefore, for this partition P , by (2.26) we have

$$U(P, f, \alpha) - L(P, f, \alpha) = \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i \leq \eta \sum_{i=1}^n \Delta \alpha_i = \eta(\alpha(b) - \alpha(a)) < \varepsilon.$$

□

Theorem 2.27. *Let f and α be monotonic increasing functions on $[a, b]$. Assume that α is continuous on $[a, b]$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$.*

Proof. If $\alpha(b) = \alpha(a)$, then the theorem is trivial. So we may assume $\alpha(b) > \alpha(a)$. Let $\varepsilon > 0$ be given and $n \in \mathbb{N}$. It is sufficient to find a partition P such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \varepsilon.$$

Put $x_0 = a$, and $x_n = b$. Recall that α is continuous on $[a, b]$ and α is monotonically increasing on $[a, b]$. Thus applying the intermediate value theorem (Theorem 1.51), we can find $x_1, \dots, x_{n-1}$ so that

$$\alpha(x_i) = \alpha(a) + i \cdot \frac{\alpha(b) - \alpha(a)}{n} \quad (i = 1, \dots, n-1)$$

and

$$x_0 = a \leq x_1 \leq \dots \leq x_{n-1} \leq b = x_n.$$

In particular, we have

$$\Delta \alpha_i := \alpha(x_i) - \alpha(x_{i-1}) = \frac{\alpha(b) - \alpha(a)}{n} \quad \forall i \in \{1, \dots, n\}.$$

Define

$$P_n := \{x_0, x_1, \dots, x_{n-1}, x_n\}.$$

Then P_n is a partition of $[a, b]$. Thus recalling f is monotonically increasing, we have

$$\begin{aligned} U(P_n, f, \alpha) - L(P_n, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \Delta \alpha_i \\ &= \sum_{i=1}^n (f(x_i) - f(x_{i-1})) \frac{\alpha(b) - \alpha(a)}{n} \\ &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^n (f(x_i) - f(x_{i-1})) \\ &= \frac{\alpha(b) - \alpha(a)}{n} (f(b) - f(a)). \end{aligned}$$

This process holds for all $n \in \mathbb{N}$. Choose a $n_0 \in \mathbb{N}$ so that

$$\frac{\alpha(b) - \alpha(a)}{n_0} (f(b) - f(a)) < \varepsilon.$$

Then finally we have

$$U(P_{n_0}, f, \alpha) - L(P_{n_0}, f, \alpha) < \varepsilon.$$

□

Exercise 2.28. Let f be a real-valued function on $[a, b]$ and α be a monotonic increasing function on $[a, b]$. Assume that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Show that the function $-f$ is also Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Corollary 2.29. Let f be a monotonically decreasing function on $[a, b]$ and α be a monotonic increasing function on $[a, b]$. Assume that α is continuous on $[a, b]$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Exercise 2.30. Prove Corollary 2.29.

Lemma 2.31. Let f be a bounded function on $[a, b]$ and α be a monotonically increasing function on $[a, b]$. Assume that there exist an open interval (c, d) and points $a_0, a_1, \dots, a_n$ such that $a_0, a_1, \dots, a_n \in [a, b] \cap (c, d)$ and

$$a_0 \leq a_1 \leq \dots \leq a_n.$$

Then

$$\sum_{i=1}^n (M_i - m_i) \Delta \alpha_i \leq (M - m)(\alpha(b \wedge d) - \alpha(a \vee c)),$$

where

$$b \wedge d := \min\{b, d\}, \quad a \vee c := \max\{a, c\}, \quad \Delta \alpha_i = \alpha(a_i) - \alpha(a_{i-1}),$$

$$M_i = \sup_{a_{i-1} \leq x \leq a_i} f(x), \quad m_i = \inf_{a_{i-1} \leq x \leq a_i} f(x), \quad (i = 1, \dots, n),$$

$$M = \sup_{a \vee c \leq x \leq b \wedge d} f(x), \quad m = \inf_{a \vee c \leq x \leq b \wedge d} f(x).$$

Proof. Note that $[a_{i-1}, a_i] \subset [a \vee c, b \wedge d]$ and thus

$$M_i - m_i \leq M - m \quad \forall i = 1, \dots, n.$$

Moreover,

$$[a_0, a_n] \subset [a, b] \cap (c, d) \subset [a \vee c, b \wedge d]$$

and thus

$$\sum_{i=1}^n \Delta \alpha_i = \alpha(a_n) - \alpha(a_0) \leq \alpha(b \wedge d) - \alpha(a \vee c)$$

since α is monotonically increasing. Therefore,

$$\sum_{i=1}^n (M_i - m_i) \Delta \alpha_i \leq (M - m) \sum_{i=1}^n \Delta \alpha_i \leq (M - m)(\alpha(b \wedge d) - \alpha(a \vee c)).$$

□

Theorem 2.32. *Let f be a bounded function on $[a, b]$ and α be a monotonically increasing function on $[a, b]$. Assume that f has only finitely many points of discontinuity on $[a, b]$ and α is continuous at every point at which f is discontinuous, i.e. there exists points $c_1, \dots, c_k \in [a, b]$ such that*

$$c_1 \leq \dots \leq c_k,$$

f is continuous at c for all $c \in [a, b] \setminus \{c_1, \dots, c_k\}$ and α is continuous at c_i for all $i = 1, \dots, k$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Proof. Let $\varepsilon > 0$ be given. For each c_i ($i = 1, \dots, k$), α is continuous at c_i and thus there exists $\delta_i > 0$ such that

$$|\alpha(x) - \alpha(c_i)| < \varepsilon \quad (2.27)$$

for all $x \in [c_i - \delta_i, c_i + \delta_i] \cap [a, b]$. Define

$$K := [a, b] \setminus \bigcup_{i=1}^k (c_i - \delta_i/2, c_i + \delta_i/2) = [a, b] \cap \left(\bigcup_{i=1}^k (c_i - \delta_i/2, c_i + \delta_i/2) \right)^c.$$

Then since K is closed and bounded, K is compact by the Heine-Borel theorem. Thus f is uniformly continuous on $K \subset [a, b]$ and there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \varepsilon \quad (2.28)$$

for all $x, y \in K \subset [a, b]$ satisfying $|x - y| < \delta$. Choose a large $n \in \mathbb{N}$ so that

$$\frac{b-a}{n} < \min\{\delta, \delta_1/2, \dots, \delta_k/2\} \quad (2.29)$$

and put

$$x_i = a + i \cdot \frac{b-a}{n} \quad (i = 0, \dots, n).$$

Define

$$P = \{x_0, \dots, x_n\} \cup \{c_1, \dots, c_k\}.$$

Then rearranging P , we can regard P as a partition, i.e. there exist $z_0, z_1, \dots, z_{n+k}$ such that

$$P = \{z_0, z_1, \dots, z_{n+k}\}$$

and

$$a = z_0 \leq z_1 \leq \dots \leq z_{n+k} = b.$$

Obviously,

$$|\Delta z_i| := |z_i - z_{i-1}| \leq \frac{b-a}{n} < \min\{\delta, \delta_1/2, \dots, \delta_k/2\}. \quad (2.30)$$

We write $i \in B_1$ iff

$$[z_{i-1}, z_i] \subset K.$$

Then

$$\{1, \dots, n+k\} = B_1 \cup (\{1, 2, \dots, n+k\} \setminus B_1).$$

and put

$$B_2 = \{1, 2, \dots, n+k\} \setminus B_1.$$

Moreover, if $i \in B_1$, then by (2.28),

$$M_i - m_i \leq \varepsilon,$$

where

$$M_i = \sup_{z_{i-1} \leq x \leq z_i} f(x), \quad \text{and} \quad m_i = \inf_{z_{i-1} \leq x \leq z_i} f(x).$$

On the other hand, if $i \in B_2$, then $[z_{i-1}, z_i]$ is not a subset of

$$K := [a, b] \cap \left(\bigcup_{i=1}^k (c_i - \delta_i/2, c_i + \delta_i/2) \right)^c,$$

thus there exists a j such that $(c_j - \delta_j/2, c_j + \delta_j/2) \cap [z_{i-1}, z_i] \neq \emptyset$. In other words, if $i \in B_2$, then $[z_{i-1}, z_i] \subset (c_j - \delta_j, c_j + \delta_j) \cap [a, b]$ by (2.30) for some $j \in \{1, \dots, k\}$. Moreover if $i_1, \dots, i_N \in B_2$, $i_1 \leq \dots \leq i_N$, and $z_{i_1-1}, z_{i_1}, z_{i_2-1}, z_{i_2}, \dots, z_{i_{N-1}-1}, z_{i_{N-1}}$ are contained in $(c_j - \delta, c_j + \delta_j) \cap [a, b]$ for some $j \in \{1, \dots, k\}$, then by Lemma 2.31 we have

$$\sum_{\ell=1}^N (M_{i_\ell} - m_{i_\ell}) \Delta \alpha_{i_\ell} \leq (M_{c_j} - m_{c_j}) (\alpha(b \wedge (c_j + \delta_j)) - \alpha(a \vee (c_j - \delta_j))),$$

where

$$M_{c_j} = \sup_{x \in [c_j - \delta_j, c_j + \delta_j] \cap [a, b]} f(x), \quad m_{c_j} = \inf_{x \in [c_j - \delta_j, c_j + \delta_j] \cap [a, b]} f(x).$$

Thus putting $M = \sup_{x \in [a, b]} |f(x)|$ and

$$\Delta \alpha_{c_j} = \alpha(b \wedge (c_j + \delta_j)) - \alpha(a \vee (c_j - \delta_j)),$$

by (2.27) we have

$$\begin{aligned} \sum_{i \in B_2} (M_i - m_i) \Delta \alpha_i &\leq \sum_{j=1}^k (M_{c_j} - m_{c_j}) \Delta \alpha_{c_j} \\ &\leq \sum_{j=1}^k 4M\varepsilon = 4kM\varepsilon. \end{aligned}$$

Therefore

$$\begin{aligned} U(P, f, \alpha) - L(P, f, \alpha) &= \sum_{i=1}^{n+k} (M_i - m_i) \Delta \alpha_i \\ &= \sum_{i \in B_1} (M_i - m_i) \Delta \alpha_i + \sum_{i \in B_2} (M_i - m_i) \Delta \alpha_i \\ &\leq \varepsilon \sum_{i \in B_1} \Delta \alpha_i + 4kM\varepsilon \\ &\leq \varepsilon (\alpha(b) - \alpha(a)) + 4kM\varepsilon = \varepsilon (\alpha(b) - \alpha(a) + 4kM). \end{aligned} \tag{2.31}$$

The theorem is proved since $\varepsilon > 0$ is an arbitrary positive constant. $\square$

Exercise 2.33. Recall that ε in (2.31) is an arbitrary positive constant. Explain why (2.31) completes the proof.

Exercise 2.34. Let f be a bounded function on $[a, b]$, α be a monotonically increasing function on $[a, b]$, A be a countable subset of $[a, b]$. Assume that A has an only one limit point in $[a, b]$, f is continuous on $[a, b] \setminus A$, and α is continuous on A . Show that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Theorem 2.35. *Let f be a Riemann-Stieltjes integrable function with respect to α on $[a, b]$. Assume that there exist $m \leq M$ such that*

$$m \leq f(x) \leq M \quad \forall x \in [a, b] \quad (2.32)$$

and continuous function ϕ on $[m, M]$. Then the composition function $h(x) := \phi(f(x))$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Proof. Let $\varepsilon > 0$ be given. Since ϕ is continuous on $[m, M]$ and $[m, M]$ is compact, ϕ is uniformly continuous on $[m, M]$. Thus there exists a $\delta > 0$ such that

$$\delta < \varepsilon \quad (2.33)$$

and

$$|\phi(s) - \phi(t)| < \varepsilon \quad \forall s, t \in [m, M] \text{ satisfying } |s - t| < \delta. \quad (2.34)$$

Since f is Riemann-Stieltjes integrable function with respect to α on $[a, b]$, there exists a partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ such that

$$U(P, f, \alpha) - L(P, f, \alpha) < \delta^2. \quad (2.35)$$

For all $i = 1, \dots, n$, put

$$M_i^f = \sup_{x_{i-1} \leq x \leq x_i} f(x), \quad m_i^f = \inf_{x_{i-1} \leq x \leq x_i} f(x), \quad M_i^h = \sup_{x_{i-1} \leq x \leq x_i} h(x),$$

and $m_i^h = \inf_{x_{i-1} \leq x \leq x_i} h(x)$. We write $i \in B_1$ iff $M_i^f - m_i^f < \delta$. Then setting $B_2 = \{1, 2, \dots, n\} \setminus B_1$, we have

$$\{1, 2, \dots, n\} = B_1 \cup B_2.$$

If $i \in B_1$, then for all $x, y \in [x_{i-1}, x_i]$

$$|f(x) - f(y)| \leq \sup_{x_{i-1} \leq x \leq x_i} f(x) - \inf_{x_{i-1} \leq x \leq x_i} f(x) = M_i^f - m_i^f < \delta$$

and thus by (2.32) and (2.34),

$$h(x) - h(y) \leq |h(x) - h(y)| = |\phi(f(x)) - \phi(f(y))| < \varepsilon,$$

which leads to

$$M_i^h - m_i^h \leq \varepsilon. \quad (2.36)$$

On the other hand, if $i \in B_2$, then $M_i^f - m_i^f \geq \delta$ and thus by (2.35)

$$\begin{aligned} \delta \sum_{i \in B_2} \Delta \alpha_i &\leq \sum_{i \in B_2} (M_i^f - m_i^f) \Delta \alpha_i \leq \sum_{i=1}^n (M_i^f - m_i^f) \Delta \alpha_i \\ &= U(P, f, \alpha) - L(P, f, \alpha) < \delta^2, \end{aligned}$$

which obviously implies

$$\sum_{i \in B_2} \Delta \alpha_i < \delta. \quad (2.37)$$

Thus defining

$$K := \sup_{m \leq t \leq M} |\phi(t)|,$$

and applying (2.36), (2.37), and (2.33), we have

$$\begin{aligned}
U(P, h, \alpha) - L(P, h, \alpha) &= \sum_{i=1}^n (M_i^h - m_i^h) \Delta \alpha_i \\
&= \sum_{i \in B_1} (M_i^h - m_i^h) \Delta \alpha_i + \sum_{i \in B_2} (M_i^h - m_i^h) \Delta \alpha_i \\
&\leq \varepsilon \sum_{i \in B_1} \Delta \alpha_i + 2K \sum_{i \in B_2} \Delta \alpha_i \\
&\leq \varepsilon \sum_{i=1}^n \Delta \alpha_i + 2K\delta \\
&\leq \varepsilon (\alpha(b) - \alpha(a) + 2K).
\end{aligned}$$

Since $\varepsilon > 0$ is an arbitrary positive constant, the theorem is proved. $\square$

Exercise 2.36. Let $A \subset \mathbb{R}$, $b \in \mathbb{R}$, $c > 0$, and $d < 0$. Prove the followings:

(i)

$$\sup(A + b) = \sup A + b,$$

where $A + b := \{a + b : a \in A\}$.

(ii)

$$\sup(cA) = c \sup A,$$

where $cA := \{ca : a \in A\}$.

(iii)

$$\sup(dA) = d \inf A.$$

(iv)

$$\inf(A + b) = \inf A + b.$$

(v)

$$\inf(cA) = c \inf A.$$

(vi)

$$\inf(dA) = d \sup A.$$

Theorem 2.37. Let f, f_1, f_2 be bounded functions on $[a, b]$ and α be a monotonically increasing function on $[a, b]$.

(i) If f_1 and f_2 are Riemann-Stieltjes integrable with respect to α on $[a, b]$, then $f_1 + f_2$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and

$$\int_a^b (f_1 + f_2) d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha. \quad (2.38)$$

(ii) If f is Riemann-Stieltjes integrable with respect to α on $[a, b]$, then for any constant $c \in \mathbb{R}$, cf is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and

$$\int_a^b cf d\alpha = c \int_a^b f d\alpha.$$

- (iii) If f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and $c \in (a, b)$, then f is Riemann-Stieltjes integrable with respect to α on $[a, c]$, Riemann-Stieltjes integrable with respect to α on $[c, b]$, and

$$\int_a^c f d\alpha + \int_c^b f d\alpha = \int_a^b f d\alpha.$$

- (iv) If f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and $\sup_{x \in [a, b]} |f(x)| \leq M$ for some positive constant $M > 0$, then

$$\left| \int_a^b f d\alpha \right| \leq M[\alpha(b) - \alpha(a)].$$

- (v) If f_1 and f_2 are Riemann-Stieltjes integrable with respect to α on $[a, b]$ and $f_1(x) \leq f_2(x)$ for all $x \in [a, b]$, then

$$\int_a^b f_1 d\alpha \leq \int_a^b f_2 d\alpha.$$

Proof. We first prove (i). Let $\varepsilon > 0$ be given. Then since f_1 and f_2 are Riemann-Stieltjes integrable with respect to α on $[a, b]$, there exist partitions P_1 and P_2 of $[a, b]$ such that

$$U(P_1, f_1, \alpha) - L(P_1, f_1, \alpha) < \frac{\varepsilon}{2}$$

and

$$U(P_2, f_2, \alpha) - L(P_2, f_2, \alpha) < \frac{\varepsilon}{2}.$$

Denote the common refinement of P_1 and P_2 by P^* . Then

$$U(P^*, f_1, \alpha) - L(P^*, f_1, \alpha) \leq U(P_1, f_1, \alpha) - L(P_1, f_1, \alpha) < \frac{\varepsilon}{2} \quad (2.39)$$

and

$$U(P^*, f_2, \alpha) - L(P^*, f_2, \alpha) \leq U(P_2, f_2, \alpha) - L(P_2, f_2, \alpha) < \frac{\varepsilon}{2}. \quad (2.40)$$

Put

$$P^* = \{x_0, x_1, \dots, x_n\}$$

and set for $i = 1, \dots, n$,

$$M_i^{f_1} = \sup_{x_{i-1} \leq x \leq x_i} f_1(x), \quad M_i^{f_2} = \sup_{x_{i-1} \leq x \leq x_i} f_2(x),$$

$$m_i^{f_1} = \inf_{x_{i-1} \leq x \leq x_i} f_1(x), \quad m_i^{f_2} = \inf_{x_{i-1} \leq x \leq x_i} f_2(x),$$

$$M_i^{f_1+f_2} = \sup_{x_{i-1} \leq x \leq x_i} (f_1(x) + f_2(x)), \quad m_i^{f_1+f_2} = \inf_{x_{i-1} \leq x \leq x_i} (f_1(x) + f_2(x)).$$

Then

$$M_i^{f_1+f_2} \leq M_i^{f_1} + M_i^{f_2}, \quad m_i^{f_1} + m_i^{f_2} \leq m_i^{f_1+f_2}. \quad (2.41)$$

Thus by (2.39) and (2.40),

$$\begin{aligned}
& U(P^*, f_1 + f_2, \alpha) - L(P^*, f_1 + f_2, \alpha) \\
&= \sum_{i=1}^n (M_i^{f_1+f_2} - m_i^{f_1+f_2}) \Delta \alpha_i \\
&\leq \sum_{i=1}^n (M_i^{f_1} + M_i^{f_2} - (m_i^{f_1} + m_i^{f_2})) \Delta \alpha_i \\
&= \sum_{i=1}^n (M_i^{f_1} - m_i^{f_1}) \Delta \alpha_i + \sum_{i=1}^n (M_i^{f_2} - m_i^{f_2}) \Delta \alpha_i \\
&= U(P^*, f_1, \alpha) - L(P^*, f_1, \alpha) + U(P^*, f_2, \alpha) - L(P^*, f_2, \alpha) < \varepsilon.
\end{aligned}$$

Therefore, $f_1 + f_2$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Next we prove (2.38). Recalling (2.41), for all partition P_1 and P_2 of $[a, b]$ and their refinement P^* , we have

$$\begin{aligned}
\int_a^b (f_1 + f_2) d\alpha &= \int_a^b (f_1 + f_2) d\alpha \leq U(P^*, f_1 + f_2, \alpha) \leq U(P^*, f_1, \alpha) + U(P^*, f_2, \alpha) \\
&\leq U(P_1, f_1, \alpha) + U(P_2, f_2, \alpha)
\end{aligned}$$

and

$$\begin{aligned}
L(P_1, f_1, \alpha) + L(P_2, f_2, \alpha) &\leq L(P^*, f_1, \alpha) + L(P^*, f_2, \alpha) \\
&\leq L(P^*, f_1 + f_2, \alpha) \\
&\leq \int_a^b (f_1 + f_2) d\alpha = \int_a^b (f_1 + f_2) d\alpha.
\end{aligned}$$

Therefore,

$$\begin{aligned}
\int_a^b (f_1 + f_2) d\alpha &\leq \inf_{P_1} U(P_1, f_1, \alpha) + \inf_{P_2} U(P_2, f_2, \alpha) \\
&= \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha = \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha
\end{aligned}$$

and

$$\begin{aligned}
\int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha &= \int_a^b f_1 d\alpha + \int_a^b f_2 d\alpha = \sup_{P_1} L(P_1, f_1, \alpha) + \sup_{P_2} L(P_2, f_2, \alpha) \\
&\leq \int_a^b (f_1 + f_2) d\alpha.
\end{aligned}$$

Second, we prove (ii). If $c = 0$, then the statement is obvious. We consider the case $c > 0$ first. Then for any partition $P = \{x_0, x_1, \dots, x_n\}$ and $i = 1, \dots, n$,

$$\sup_{x_{i-1} \leq x \leq x_i} cf(x) = c \sup_{x_{i-1} \leq x \leq x_i} f(x)$$

and

$$\inf_{x_{i-1} \leq x \leq x_i} cf(x) = c \inf_{x_{i-1} \leq x \leq x_i} f(x).$$

Thus

$$\int_a^{\bar{b}} c f d\alpha = c \int_a^{\bar{b}} f d\alpha = c \int_a^b f d\alpha$$

and

$$\int_a^{\underline{b}} c f d\alpha = c \int_a^{\underline{b}} f d\alpha = c \int_a^b f d\alpha.$$

On the other hand, if $c < 0$, then for any partition $P = \{x_0, x_1, \dots, x_n\}$ and $i = 1, \dots, n$,

$$\sup_{x_{i-1} \leq x \leq x_i} c f(x) = c \inf_{x_{i-1} \leq x \leq x_i} f(x)$$

and

$$\inf_{x_{i-1} \leq x \leq x_i} c f(x) = c \sup_{x_{i-1} \leq x \leq x_i} f(x).$$

Thus

$$\int_a^{\bar{b}} c f d\alpha = c \int_a^{\underline{b}} f d\alpha = c \int_a^b f d\alpha$$

and

$$\int_a^{\underline{b}} c f d\alpha = c \int_a^{\bar{b}} f d\alpha = c \int_a^b f d\alpha.$$

We left the proof of (iii), (iv), and (v) as an exercise. $\square$

Exercise 2.38. Prove (iii)-(v) in Theorem 2.37.

Exercise 2.39 (Linear property of the Riemann-Stieltjes integral). Let f_1 and f_2 be Riemann-Stieltjes integrable functions with respect to α on $[a, b]$ and $c_1, c_2 \in \mathbb{R}$. Show that the function $c_1 f_1 + c_2 f_2$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$ and

$$\int_a^b (c_1 f_1 + c_2 f_2) d\alpha = c_1 \int_a^b f_1 d\alpha + c_2 \int_a^b f_2 d\alpha.$$

Theorem 2.40. Let f be a bounded functions on $[a, b]$ and $\alpha, \alpha_1, \alpha_2$ be monotonically increasing functions on $[a, b]$.

- (i) If f is Riemann-Stieltjes integrable with respect to α_1 on $[a, b]$ and f is Riemann-Stieltjes integrable with respect to α_2 on $[a, b]$, then f is Riemann-Stieltjes integrable with respect to $\alpha_1 + \alpha_2$ on $[a, b]$ and

$$\int_a^b f d(\alpha_1 + \alpha_2) = \int_a^b f d\alpha_1 + \int_a^b f d\alpha_2.$$

- (ii) If f is Riemann-Stieltjes integrable with respect to α on $[a, b]$, then for any positive constant c , f is Riemann-Stieltjes integrable with respect to $c\alpha$ on $[a, b]$ and

$$\int_a^b f d(c\alpha) = c \int_a^b f d\alpha.$$

Proof. We skip the proof of this theorem since the proof is quite similar to that of Theorem 2.37. $\square$

Exercise 2.41. Prove Theorem 2.40.

Theorem 2.42. Let f and g be bounded functions on $[a, b]$. Assume that f and g are Riemann-Stieltjes integrable functions with respect to α on $[a, b]$. Then

- (i) fg is Riemann-Stieltjes integrable functions with respect to α on $[a, b]$.
- (ii) $|f|$ is Riemann-Stieltjes integrable functions with respect to α on $[a, b]$ and

$$\left| \int_a^b f d\alpha \right| \leq \int_a^b |f| d\alpha. \quad (2.42)$$

Proof. First we prove (i). By Theorem 2.37 (i) and (ii), $f+g$ and $f-g$ are Riemann-Stieltjes integrable functions with respect to α on $[a, b]$. Set $\phi(t) = t^2$. Then by Theorem 2.35, $\phi(f+g) = (f+g)^2$ and $\phi(f-g) = (f-g)^2$ are Riemann-Stieltjes integrable functions with respect to α on $[a, b]$. Thus by Theorem 2.37 (i) and (ii) again,

$$fg = \frac{1}{4} (4fg) = \frac{1}{4} ((f+g)^2 - (f-g)^2)$$

is a Riemann-Stieltjes integrable function with respect to α on $[a, b]$. Next we prove (ii). Set $\phi(t) = |t|$. Then Theorem 2.35 shows that $|f|$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Thus it only remains to show (2.42). Choose $c = 1$ or $c = -1$ so that

$$\left| \int_a^b f d\alpha \right| = c \int_a^b f d\alpha.$$

Then

$$cf(x) \leq |f(x)| \quad \forall x \in [a, b]$$

and by Theorem 2.37,

$$\begin{aligned} \left| \int_a^b f d\alpha \right| &= c \int_a^b f d\alpha = \int_a^b cf d\alpha \\ &\leq \int_a^b |f| d\alpha. \end{aligned}$$

$\square$

Exercise 2.43. Let f and g be bounded functions on $[a, b]$. Assume that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Prove or disprove that fg is Riemann-Stieltjes integrable with respect to α on $[a, b]$.

Definition 2.44. The unit step function I is defined by

$$I(x) = \begin{cases} 0 & (x \leq 0) \\ 1 & (x > 0). \end{cases}$$

Obviously, for any $s \in \mathbb{R}$,

$$I(x-s) = \begin{cases} 0 & (x \leq s) \\ 1 & (x > s). \end{cases}$$

and for all $a \leq b$,

$$I(b-x) - I(a-x) = \begin{cases} 0 & (b \leq x) \\ 1 & (b > x) \end{cases} - \begin{cases} 0 & (a \leq x) \\ 1 & (a > x) \end{cases} = \begin{cases} 1 & (x \in [a, b)) \\ 0 & (x \notin [a, b)). \end{cases}$$

More generally, for any set A , we define the indicator function $I_A(x)$ as

$$I_A(x) = \begin{cases} 1 & (x \in A) \\ 0 & (x \in A^c). \end{cases}$$

In particular,

$$I(b-x) - I(a-x) = I_{[a,b)}(x).$$

Theorem 2.45. Let $a < s < b$, f be a bounded function on $[a, b]$, and $\alpha(x) = I(x-s)$. Assume that f is continuous at s . Then

$$\int_a^b f d\alpha = f(s).$$

Proof. For all $n \in \mathbb{N}$, set

$$x_i^n = a + i \frac{b-a}{n} \quad (i = 0, \dots, n)$$

and

$$P_n = \{x_0^n, x_1^n, \dots, x_n^n\}.$$

Then P_n is a partition of $[a, b]$. Since $s \in (a, b)$, for each $n \in \mathbb{N}$, there exists a $k \in \{1, \dots, n\}$ such that

$$s \in [x_{k-1}^n, x_k^n).$$

Thus for each n , we have

$$\Delta\alpha_k = \alpha(x_k^n) - \alpha(x_{k-1}^n) = I(x_k^n - s) - I(x_{k-1}^n - s) = I_{[x_{k-1}^n, x_k^n)}(s) = 1.$$

$$\Delta\alpha_i = \alpha(x_i^n) - \alpha(x_{i-1}^n) = I(x_i^n - s) - I(x_{i-1}^n - s) = 0$$

for all $i \in \{1, \dots, n\} \setminus \{k\}$, and

$$\begin{aligned} U(P_n, f, \alpha) &= \sum_{i=1}^n M_i \Delta\alpha_i \\ &= M_k = \sup_{x \in [x_{k-1}^n, x_k^n]} f(x), \end{aligned} \quad (2.43)$$

$$\begin{aligned} L(P_n, f, \alpha) &= \sum_{i=1}^n m_i \Delta\alpha_i \\ &= m_k = \inf_{x \in [x_{k-1}^n, x_k^n]} f(x), \end{aligned} \quad (2.44)$$

and

$$\begin{aligned} U(P_n, f, \alpha) - L(P_n, f, \alpha) &= \sum_{i=1}^n (M_i - m_i) \Delta\alpha_i \\ &= M_k - m_k = \sup_{x \in [x_{k-1}^n, x_k^n]} f(x) - \inf_{x \in [x_{k-1}^n, x_k^n]} f(x). \end{aligned} \quad (2.45)$$

Let $\varepsilon > 0$ be given. Then since f is continuous at s , there exists a $\delta > 0$ such that

$$|f(x) - f(s)| < \frac{\varepsilon}{3} \quad (2.46)$$

for all $x \in (s - \delta, s + \delta)$. Take $n_0 \in \mathbb{N}$ large enough so that

$$\frac{b-a}{n_0} < \delta.$$

Recall that for this n_0 , there exists a $k_0 \in \{1, \dots, n_0\}$ such that

$$s \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]$$

Since $x_{k_0}^{n_0} - x_{k_0-1}^{n_0} = \frac{b-a}{n_0} < \delta$, we have

$$[x_{k_0-1}^{n_0}, x_{k_0}^{n_0}] \subset (s - \delta, s + \delta).$$

Thus applying (2.46) and the triangle inequality for all $x, y \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]$,

$$f(x) - f(y) \leq |f(x) - f(y)| \leq |f(x) - f(s)| + |f(y) - f(s)| < \frac{2\varepsilon}{3}.$$

In particular,

$$\sup_{x \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]} f(x) - f(s) \leq \frac{2\varepsilon}{3} < \varepsilon, \quad (2.47)$$

$$f(s) \leq \inf_{x \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]} f(x) + \frac{2\varepsilon}{3} < \inf_{x \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]} f(x) + \varepsilon, \quad (2.48)$$

and

$$\sup_{x \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]} f(x) - \inf_{x \in [x_{k_0-1}^{n_0}, x_{k_0}^{n_0}]} f(x) \leq \frac{2\varepsilon}{3} < \varepsilon.$$

Therefore by (2.45),

$$U(P_{n_0}, f, \alpha) - L(P_{n_0}, f, \alpha) < \varepsilon,$$

which implies that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Applying (2.43), (2.44), (2.47), and (2.48), we have

$$f(s) - \varepsilon < m_{k_0}^{n_0} = L(P_{n_0}, f, \alpha) \leq \int_a^b f d\alpha = \int_a^b f d\alpha$$

and

$$\int_a^b f d\alpha = \int_a^{\bar{b}} f d\alpha \leq U(P_{n_0}, f, \alpha) = M_{k_0}^{n_0} < f(s) + \varepsilon.$$

Since $\varepsilon > 0$ is an arbitrary positive constant, we have

$$\int_a^b f d\alpha = f(s).$$

□

Exercise 2.46. Let a_n and b_n be sequences in $\mathbb{R}$. Assume that

$$a_n \geq 0, \quad |b_n| \leq a_n \quad \forall n \in \mathbb{N},$$

and $\sum_{n=1}^{\infty} a_n$ converges. Show that the series $\sum_{n=1}^{\infty} b_n$ absolutely converges, i.e. the sequence $\sum_{n=1}^k |b_n|$ converges to a real number as $k \rightarrow \infty$.

Theorem 2.47. Let c_n be a nonnegative sequence, s_n be a sequence of distinct points in (a, b) , and

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n),$$

i.e. $c_n \geq 0$, $s_n \in (a, b)$ for all $n \in \mathbb{N}$ and $s_j \neq s_k$ for all $j \neq k$. Assume that f is continuous on $[a, b]$ and $\sum_{n=1}^{\infty} c_n$ converges. Then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

Proof. For each $x \in [a, b]$, $c_n I(x - s_n) \leq c_n$ for all $n \in \mathbb{N}$ and thus $\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$ is well-defined. Similarly, $\sum_{n=1}^{\infty} c_n f(s_n)$ is well-defined since

$$|c_n f(s_n)| \leq M c_n,$$

where

$$M := \sup_{x \in [a, b]} |f(x)| < \infty$$

since f is continuous on $[a, b]$ and $[a, b]$ is a compact set. Moreover, since each $c_n I(x - s_n)$ is monotonically increasing, α is also monotonically increasing. Thus since f is continuous on $[a, b]$, f is Riemann-Stieltjes integrable with respect to α on $[a, b]$. Let $\varepsilon > 0$ be given. Choose a $N \in \mathbb{N}$ so that

$$\sum_{n=N+1}^{\infty} c_n := \sum_{n=1}^{\infty} c_n - \sum_{n=1}^N c_n < \varepsilon. \quad (2.49)$$

Put

$$\alpha_1(x) := \sum_{n=1}^N c_n I(x - s_n), \quad \text{and} \quad \alpha_2(x) := \alpha(x) - \alpha_1(x) = \sum_{n=N+1}^{\infty} c_n I(x - s_n).$$

Obviously both α_1 and α_2 are monotonically increasing on $[a, b]$. Thus f is Riemann-Stieltjes integrable with respect to α_1 and α_2 on $[a, b]$, respectively since f is continuous on $[a, b]$. Observe that

$$\begin{aligned} \alpha(b) &= \sum_{n=1}^{\infty} c_n I(b - s_n) = \sum_{n=1}^{\infty} c_n, \\ \alpha_1(b) &= \sum_{n=1}^N c_n I(b - s_n) = \sum_{n=1}^N c_n \\ \alpha(a) &= \sum_{n=1}^{\infty} c_n I(a - s_n) = 0, \quad \alpha_1(a) = 0, \end{aligned}$$

and by (2.49)

$$\alpha_2(b) - \alpha_2(a) = \alpha(b) - \alpha_1(b) - \alpha(a) + \alpha_1(a) = \sum_{n=N+1}^{\infty} c_n < \varepsilon.$$

Thus applying Theorem 2.37(iv), we have

$$\left| \int_a^b f d\alpha_2 \right| \leq M(\alpha_2(b) - \alpha_2(a)) < M\varepsilon. \quad (2.50)$$

On the other hand, by Theorem 2.40 and Theorem 2.45,

$$\begin{aligned}\int_a^b f d\alpha_1 &= \int_a^b f d\left(\sum_{n=1}^N c_n I(x - s_n)\right) \\ &= \sum_{n=1}^N c_n \int_a^b f d(I(x - s_n)) \\ &= \sum_{n=1}^N c_n f(s_n).\end{aligned}$$

Therefore recalling (2.50) and applying Theorem 2.40 and Theorem 2.45 again, we have

$$\begin{aligned}\left|\int_a^b f d\alpha - \sum_{n=1}^{\infty} c_n f(s_n)\right| &= \left|\int_a^b f d\alpha_1 + \int_a^b f d\alpha_2 - \sum_{n=1}^N c_n f(s_n) - \sum_{n=N+1}^{\infty} c_n f(s_n)\right| \\ &= \left|\int_a^b f d\alpha_2 - \sum_{n=N+1}^{\infty} c_n f(s_n)\right| \leq 2M\varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ is an arbitrary positive constant,

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

□

Lemma 2.48. *Let f be a bounded function on $[a, b]$, α be a monotonically increasing function, and P be a partition of $[a, b]$. Then*

$$\int_a^b f d\alpha = \inf_{P^* \supset P} U(P^*, f, \alpha) := \inf\{U(P^*, f, \alpha) : P^* \text{ is a refinement of } P\}.$$

Proof. Obviously,

$$\int_a^b f d\alpha \leq \inf_{P^* \supset P} U(P^*, f, \alpha).$$

On the other hand, let P_1 be a partition of $[a, b]$. Then $P_1 \cup P$ becomes a refinement of P and

$$\inf_{P^* \supset P} U(P^*, f, \alpha) \leq U(P_1 \cup P, f, \alpha) \leq U(P_1, f, \alpha).$$

Since P_1 is an arbitrary partition of $[a, b]$, we have

$$\inf_{P^* \supset P} U(P^*, f, \alpha) \leq \int_a^b f d\alpha.$$

□

Lemma 2.49. *Let f be a bounded function on $[a, b]$ and α be a monotonically increasing function. Assume that α is differentiable on $[a, b]$ and its derivative α' is Riemann integrable on $[a, b]$. Then*

(i)

$$\int_a^b f d\alpha = \int_a^b f \alpha' dx$$

(ii)

$$\int_a^b f d\alpha = \int_a^b f \alpha' dx.$$

Proof. Because of similarity of the proof, we only prove (i). Let $\varepsilon > 0$ be given. Since α' is Riemann integrable on $[a, b]$, there exists a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that

$$U(P, \alpha') - L(P, \alpha') < \varepsilon \quad (2.51)$$

By the mean-value theorem, for each $i \in \{1, \dots, n\}$, there exists $t_i \in [x_{i-1}, x_i]$ such that

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)\Delta x_i.$$

Moreover, set

$$M_i^{\alpha'} = \sup_{x \in [x_{i-1}, x_i]} \alpha'(x) \quad \text{and} \quad m_i^{\alpha'} = \inf_{x \in [x_{i-1}, x_i]} \alpha'(x)$$

Then due to (2.51), for all i and $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^n |\alpha'(s_i) - \alpha'(t_i)| \Delta x_i \leq \sum_{i=1}^n (M_i^{\alpha'} - m_i^{\alpha'}) \Delta x_i = U(P, \alpha') - L(P, \alpha') < \varepsilon.$$

Thus putting

$$M := \sup_{x \in [a, b]} |f(x)|,$$

for all $s_i \in [x_{i-1}, x_i]$ we have

$$\begin{aligned} \left| \sum_{i=1}^n f(s_i) \Delta\alpha_i - \sum_{i=1}^n f(s_i) \alpha'(s_i) \Delta x_i \right| &= \left| \sum_{i=1}^n f(s_i) \alpha'(t_i) \Delta x_i - \sum_{i=1}^n f(s_i) \alpha'(s_i) \Delta x_i \right| \\ &\leq \sum_{i=1}^n |f(s_i)| |\alpha'(t_i) - \alpha'(s_i)| \Delta x_i \\ &\leq M \sum_{i=1}^n |\alpha'(t_i) - \alpha'(s_i)| \Delta x_i < M\varepsilon. \end{aligned} \quad (2.52)$$

In particular, for all $s_i \in [x_{i-1}, x_i]$,

$$\begin{aligned} \sum_{i=1}^n f(s_i) \Delta\alpha_i &\leq \sum_{i=1}^n f(s_i) \alpha'(s_i) \Delta x_i + M\varepsilon \\ &\leq U(P, f\alpha') + M\varepsilon. \end{aligned} \quad (2.53)$$

Thus

$$U(P, f, \alpha) \leq U(P, f\alpha') + M\varepsilon \quad (2.54)$$

Similarly,

$$U(P, f\alpha') \leq U(P, f, \alpha) + M\varepsilon. \quad (2.55)$$

Note that for any refinement P^* of P , (2.54) and (2.55) hold since

$$U(P^*, \alpha') - L(P^*, \alpha') \leq U(P, \alpha') - L(P, \alpha') < \varepsilon.$$

In other words, we have

$$\int_a^b f d\alpha \leq U(P^*, f, \alpha) \leq U(P^*, f\alpha') + M\varepsilon$$

Similarly,

$$\int_a^b f\alpha' dx \leq U(P^*, f\alpha') \leq U(P^*, f, \alpha) + M\varepsilon.$$

for all refinement P^* of P . Finally by Lemma 2.48, we have

$$\int_a^b f d\alpha \leq \int_a^b f\alpha' dx + M\varepsilon$$

Similarly,

$$\int_a^b f\alpha' dx \leq \int_a^b f d\alpha + M\varepsilon.$$

Since $\varepsilon > 0$ is an arbitrary positive constant, we conclude

$$\int_a^b f\alpha' dx = \int_a^b f d\alpha.$$

□

Exercise 2.50. Prove (ii) in Lemma 2.49.

Exercise 2.51. Prove that (2.53) implies (2.54).

Exercise 2.52. Prove (2.55) by using (2.52).

Theorem 2.53. Let f be a bounded function on $[a, b]$ and α be a monotonically increasing function. Assume that α is differentiable on $[a, b]$ and its derivative α' is Riemann integrable on $[a, b]$. Then f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ if and only if $f\alpha'$ is Riemann integrable on $[a, b]$.

Corollary 2.54. Let f be a bounded function on $[a, b]$ and α be a monotonically increasing differentiable function on $[a, b]$ whose derivative α' is Riemann integrable on $[a, b]$. Assume that f is Riemann-Stieltjes integrable with respect to α on $[a, b]$ or $f\alpha'$ is Riemann integrable on $[a, b]$. Then

$$\int_a^b f d\alpha = \int_a^b f\alpha' dx =: \int_a^b f(x)\alpha'(x)dx.$$

Exercise 2.55. Prove Theorem 2.53.

Exercise 2.56. Prove Corollary 2.54.

Theorem 2.57. Let φ be a strictly increasing continuous function that maps an interval $[A, B]$ onto $[a, b]$, α be a monotonically increasing function on $[a, b]$, and f be a Riemann-Stieltjes integrable function with respect to α on $[a, b]$. Define

$$\beta(y) = \alpha(\varphi(y)) \quad \text{and} \quad g(y) = f(\varphi(y)) \quad (y \in [A, B]).$$

Then g is Riemann-Stieltjes integrable with respect to β on $[A, B]$ and

$$\int_A^B g d\beta = \int_a^b f d\alpha.$$

Proof. Note that φ is 1-1 function from $[A, B]$ onto $[a, b]$ since φ is strictly increasing. Thus there exists the inverse function φ^{-1} from $[a, b]$ onto $[A, B]$. For any partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$, there exists a corresponding partition $Q = \{y_0, \dots, y_n\}$ of $[A, B]$ such that

$$y_i = \varphi^{-1}(x_i) \quad \text{and} \quad x_i = \varphi(y_i) \quad \forall i \in \{0, 1, \dots, n\}.$$

Similarly, for any partition $Q = \{y_0, \dots, y_n\}$ of $[A, B]$, there exists a corresponding partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that

$$y_i = \varphi^{-1}(x_i) \quad \text{and} \quad x_i = \varphi(y_i) \quad \forall i \in \{0, 1, \dots, n\}.$$

Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$ and $Q = \{y_0, \dots, y_n\}$ be the corresponding partition of $[A, B]$. For $i = 1, \dots, n$, set

$$\begin{aligned} M_i^f &:= \sup_{x \in [x_{i-1}, x_i]} f(x), & M_i^g &:= \sup_{y \in [y_{i-1}, y_i]} g(y), \\ m_i^f &:= \inf_{x \in [x_{i-1}, x_i]} f(x), & m_i^g &:= \inf_{y \in [y_{i-1}, y_i]} g(y). \end{aligned}$$

Observe that for each i ,

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha(\varphi(y_i)) - \alpha(\varphi(y_{i-1})) = \beta(y_i) - \beta(y_{i-1}) = \Delta\beta_i.$$

Moreover for all $i = 1, \dots, n$ and $x \in [x_{i-1}, x_i]$,

$$\varphi^{-1}(x) \in [y_{i-1}, y_i]$$

and

$$f(x) = f(\varphi(\varphi^{-1}(x))) = g(\varphi^{-1}(x)),$$

which lead to

$$M_i^f = M_i^g, \quad \text{and} \quad m_i^f = m_i^g.$$

Therefore

$$U(P, f, \alpha) = \sum_{i=1}^n M_i^f \Delta\alpha_i = \sum_{i=1}^n M_i^g \Delta\beta_i = U(Q, g, \beta)$$

and

$$L(P, f, \alpha) = \sum_{i=1}^n m_i^f \Delta\alpha_i = \sum_{i=1}^n m_i^g \Delta\beta_i = L(Q, g, \beta).$$

Let $\varepsilon > 0$ be given. Since f is Riemann-integrable with respect to α on $[a, b]$, there exists a partition P_0 such that

$$U(P_0, f, \alpha) - L(P_0, f, \alpha) < \varepsilon.$$

Considering the corresponding partition Q_0 of $[A, B]$, we have

$$U(Q_0, g, \beta) - L(Q_0, g, \beta) = U(P_0, f, \alpha) - L(P_0, f, \alpha) < \varepsilon.$$

Thus g is Riemann-Stieltjes integrable with respect to β on $[A, B]$. Moreover,

$$\begin{aligned} \int_a^b f d\alpha &= \int_a^b f d\alpha := \inf\{U(P, f, \alpha) : P \text{ is a partition of } [a, b]\} \\ &= \inf\{U(Q, g, \beta) : P \text{ is a partition of } [a, b]\} \\ &= \inf\{U(Q, g, \beta) : Q \text{ is a partition of } [A, B]\} \\ &= \int_A^B g d\beta = \int_A^B g d\beta. \end{aligned}$$

□

Corollary 2.58 (Change of variable). *Let φ be a strictly increasing differentiable function that maps an interval $[A, B]$ onto $[a, b]$, and f be a Riemann integrable function on $[a, b]$. Assume that φ' is Riemann integrable on $[A, B]$. Then*

$$\int_a^b f(x)dx = \int_A^B f(\varphi(y))\varphi'(y)dy.$$

Proof. Set $\alpha(x) = x$ and recall

$$\beta(y) = \alpha(\varphi(y)) \quad \text{and} \quad g(y) = f(\varphi(y)) \quad (y \in [a, b]).$$

Then $\beta(y) = \varphi(y)$ and by Theorem 2.57,

$$\int_A^B f(\varphi(y))d\varphi(y) = \int_A^B g d\beta = \int_a^b f d\alpha = \int_a^b f dx.$$

Finally applying Corollary 2.54, we have

$$\int_A^B f(\varphi(y))\varphi'(y)dy = \int_a^b f dx.$$

□

Theorem 2.59. *Let f be a Riemann integrable function on $[a, b]$. Define*

$$F(x) = \int_a^x f(t)dt \quad (a \leq x \leq b).$$

Then F is continuous on $[a, b]$.

Proof. Since f is Riemann integrable on $[a, b]$, f is bounded on $[a, b]$. Thus there exists a positive constant M such that

$$|f(t)| \leq M \quad \forall t \in [a, b].$$

Let $a \leq x \leq y \leq b$. Then by Theorem 2.37,

$$|F(y) - F(x)| = \left| \int_a^y f(t)dt - \int_a^x f(t)dt \right| = \left| \int_x^y f(t)dt \right| \leq M(y - x). \quad (2.56)$$

Let $\varepsilon > 0$ be given. Take $\delta = \frac{\varepsilon}{M}$. If $|x - y| < \delta$, then by (2.56)

$$|F(x) - F(y)| \leq M|x - y| < M\delta = \varepsilon.$$

Therefore F is uniformly continuous on $[a, b]$.

□

Exercise 2.60. *Due to Exercise 2.10, if f is Riemann integrable on $[a, b]$, then f is bounded. Prove or disprove that if f is Riemann-Stieltjes integrable with respect to α on $[a, b]$, then f is bounded.*

Theorem 2.61. *Let f be a Riemann integrable function on $[a, b]$. Define*

$$F(x) = \int_a^x f(t)dt \quad (a \leq x \leq b).$$

Assume that f is continuous at $x_0 \in [a, b]$. Then F is differentiable at x_0 and

$$F'(x_0) = f(x_0).$$

Proof. Let $\varepsilon > 0$ be given. Since f is continuous at x_0 , there exists a $\delta > 0$ such that

$$|f(x) - f(x_0)| < \frac{\varepsilon}{2} \quad (2.57)$$

for all $x \in (x_0 - \delta, x_0 + \delta) \cap [a, b]$. Note that by the linear property of the Riemann-Stieltjes integral, for all $a \leq x < x_0 < y \leq b$

$$f(x_0) = f(x_0) \frac{y - x_0}{y - x_0} = f(x_0) \frac{\int_{x_0}^y 1 dt}{y - x_0} = \frac{1}{y - x_0} \int_{x_0}^y f(x_0) dt$$

and

$$f(x_0) = f(x_0) \frac{x_0 - x}{x_0 - x} = f(x_0) \frac{\int_x^{x_0} 1 dt}{x_0 - x} = \frac{1}{x_0 - x} \int_x^{x_0} f(x_0) dt$$

Thus by the linear property again, if $y \in (x_0, x_0 + \delta) \cap [a, b]$, then

$$\begin{aligned} \left| \frac{F(y) - F(x_0)}{y - x_0} - f(x_0) \right| &= \left| \frac{\int_{x_0}^y f(t) dt}{y - x_0} - \frac{1}{y - x_0} \int_{x_0}^y f(x_0) dt \right| \\ &= \left| \frac{1}{y - x_0} \int_{x_0}^y (f(t) - f(x_0)) dt \right|. \end{aligned}$$

Therefore by (2.57) and Theorem 2.37(iv), we have

$$\left| \frac{F(y) - F(x_0)}{y - x_0} - f(x_0) \right| \leq \frac{1}{y - x_0} \frac{\varepsilon}{2} (y - x_0) = \frac{\varepsilon}{2} < \varepsilon.$$

Similarly, for all $x \in (x_0 - \delta, x_0) \cap [a, b]$, we have

$$\left| \frac{F(x) - F(x_0)}{x - x_0} - f(x_0) \right| < \varepsilon. \quad (2.58)$$

Thus F is differentiable at x_0 and

$$F'(x_0) = f(x_0).$$

□

Exercise 2.62. Prove (2.58).

Exercise 2.63. Let f be a continuous function on $[a, b]$. Prove or disprove that there exists a differentiable function F on $[a, b]$ such that

$$F'(x) = f(x) \quad \forall x \in [a, b].$$

Theorem 2.64 (The fundamental theorem of calculus). *Let f be a Riemann integrable function on $[a, b]$. Assume that there exists a differentiable function F on $[a, b]$ such that $F'(x) = f(x)$ for all $x \in [a, b]$. Then*

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof. Let $\varepsilon > 0$ be given. Then there exists a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that

$$U(P, f) - L(P, f) < \varepsilon.$$

By the mean-value theorem, there exist $\tilde{t}_i \in [x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = f(\tilde{t}_i) \Delta x_i \quad (i = 1, \dots, n).$$

Thus

$$\sum_{i=1}^n f(\tilde{t}_i) \Delta x_i = F(x_n) - F(x_0) = F(b) - F(a).$$

Due to Theorem 2.20(iii), for all $t_i \in [x_{i-1}, x_i]$ ($i = 1, \dots, n$),

$$\left| \sum_{i=1}^n f(t_i) \Delta x_i - \int_a^b f dx \right| < \varepsilon.$$

In particular, we have

$$\left| F(b) - F(a) - \int_a^b f dx \right| = \left| \sum_{i=1}^n f(\tilde{t}_i) \Delta x_i - \int_a^b f dx \right| < \varepsilon.$$

Since $\varepsilon > 0$ is an arbitrary positive constant, the theorem is proved. $\square$

Theorem 2.65 (Integration by parts). *Let F and G be differentiable function on $[a, b]$. Assume that F' and G' are Riemann integrable on $[a, b]$. Then*

$$\int_a^b F(x)G'(x)dx = F(b)G(b) - F(a)G(a) - \int_a^b F'(x)G(x)dx.$$

Proof. Set

$$H(x) = F(x)G(x).$$

Then by the product rule of differentiation,

$$H'(x) = F'(x)G(x) + F(x)G'(x).$$

Since G and F are differentiable on $[a, b]$, they are also continuous on $[a, b]$. Thus by Theorem 2.26, G and F are Riemann integrable on $[a, b]$. Since we assumed that $F'(x)$ and $G'(x)$ are Riemann integrable on $[a, b]$, $F'(x)G(x)$, $F(x)G'(x)$, $H'(x)$ are Riemann integrable by Theorem 2.42 and the linearity. Therefore applying the fundamental theorem of calculus,

$$\int_a^b H'(x)dx = H(b) - H(a),$$

which is equivalent to

$$\int_a^b (F'(x)G(x) + F(x)G'(x))dx = F(b)G(b) - F(a)G(a).$$

Finally applying the linear property of the Riemann integral, the theorem is proved. $\square$

3. SEQUENCE AND SERIES OF FUNCTIONS

Definition 3.1. *Let $E \subset \mathbb{R}$ and $\{f_n : n \in \mathbb{N}\}$ be a sequence of real-valued functions on E , i.e. for each $n \in \mathbb{N}$, f_n is a function from E to $\mathbb{R}$. Assume that for each $x \in E$, the sequence $\{f_n(x)\}$ converges and define*

$$f(x) := \lim_{n \rightarrow \infty} f_n(x).$$

*Then we say that f_n converges to f pointwise on E . Moreover, f is called **the limit of f_n** , **the limit function of f_n** , or **the pointwise limit of f_n** .*

It is very natural to ask if this pointwise limit of functions can preserve the properties of functions. Here is a list of questions.

- Assume that f_n converges to f pointwise on $[a, b]$ and f_n is continuous on $[a, b]$ for all n . Then is $f(x)$ continuous on $[a, b]$? In other words, for any $c \in E$,

$$\lim_{x \rightarrow c} \lim_{n \rightarrow \infty} f_n(x) = \lim_{x \rightarrow c} f(x) = f(c) = \lim_{n \rightarrow \infty} f_n(c) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow c} f_n(x)?$$

(the limit is considered as the left limit or right limit at the end points of $[a, b]$).

- Assume that f_n converges to f pointwise on $[a, b]$ and f_n is differentiable on $[a, b]$ for all n . Then is $f(x)$ differentiable on $[a, b]$ and

$$\begin{aligned} f'(c) &= \lim_{x \rightarrow c} \frac{f(x) - f(c)}{x - c} = \lim_{x \rightarrow c} \lim_{n \rightarrow \infty} \frac{f_n(x) - f_n(c)}{x - c} = \lim_{n \rightarrow \infty} \lim_{x \rightarrow c} \frac{f_n(x) - f_n(c)}{x - c} \\ &= \lim_{n \rightarrow \infty} f'_n(c)? \end{aligned}$$

- Assume that f_n converges to f pointwise on $[a, b]$ and f_n is Riemann integrable on $[a, b]$ for all n . Then is $f(x)$ Riemann integrable on $[a, b]$ and

$$\int_a^b f(x) dx = \int_a^b \lim_{n \rightarrow \infty} f_n(x) dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) dx?$$

All answers to above questions are “No”. We can easily find counter examples as follows.

Example 3.2. For all $n \in \mathbb{N}$, define

$$f_n(x) = \begin{cases} \frac{1}{1+nx} & (x \neq -\frac{1}{n}) \\ 0 & (x = -\frac{1}{n}). \end{cases}$$

Then

$$\lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0 & (x \neq 0) \\ 1 & (x = 0). \end{cases}$$

In particular,

$$\lim_{x \rightarrow 0} \lim_{n \rightarrow \infty} f_n(x) = 0.$$

On the other hand, since for each n , f_n is continuous at 0,

$$1 = \lim_{n \rightarrow \infty} f_n(0) = \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} f_n(x).$$

Therefore

$$\lim_{x \rightarrow 0} \lim_{n \rightarrow \infty} f_n(x) \neq \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} f_n(x).$$

Moreover,

$$f(x) = \lim_{n \rightarrow \infty} f_n(x)$$

is not differentiable at 0 even though f_n is differentiable at 0 for all n .

Next example shows that

$$f'(x) \neq \lim_{n \rightarrow \infty} f'_n(x)$$

even though $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ is differentiable on $\mathbb{R}$.

Example 3.3. For all $n \in \mathbb{N}$, define

$$f_n(x) = \frac{\sin(nx)}{\sqrt{n}} \quad (x \in \mathbb{R}).$$

Then for all $x \in \mathbb{R}$,

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) = 0.$$

Thus f is differentiable on $\mathbb{R}$ and $f'(x) = 0$ for all $x \in \mathbb{R}$. On the other hand,

$$f'_n(x) = n \frac{\cos(nx)}{\sqrt{n}} = \sqrt{n} \cos(nx)$$

and thus

$$\lim_{n \rightarrow \infty} f'_n(0) = \lim_{n \rightarrow \infty} \sqrt{n} = \infty.$$

Therefore

$$f'(0) \neq \lim_{n \rightarrow \infty} f'_n(0).$$

Exercise 3.4. Let $a < b$ and f_n be a sequence of real-valued differentiable functions on (a, b) . Assume that

$$\lim_{n \rightarrow \infty} f_n(x) \text{ and } \lim_{n \rightarrow \infty} f'_n(x)$$

exist for all $x \in (a, b)$, and the function $f(x) := \lim_{n \rightarrow \infty} f_n(x)$ is differentiable on (a, b) . Prove or disprove that

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad \forall x \in (a, b).$$

Example 3.5. For all $n \in \mathbb{N}$, define

$$f_n(x) = n^2 x(1 - x^2)^n \quad (x \in \mathbb{R}).$$

Then for each n , f_n is Riemann integrable on $[0, 1]$ since f is continuous on $[0, 1]$. Observe that for $a \in (0, 1)$, by L'Hôpital's rule,

$$\lim_{x \rightarrow \infty} x^2 a^x = \lim_{x \rightarrow \infty} \frac{x^2}{a^{-x}} = \lim_{x \rightarrow \infty} \frac{2x}{-a^{-x} \ln a} = \lim_{x \rightarrow \infty} \frac{2}{a^{-x} (\ln a)^2} = \lim_{x \rightarrow \infty} a^x \frac{2}{(\ln a)^2} = 0$$

and obviously

$$\lim_{x \rightarrow \infty} x^2 \cdot 0^x = 0.$$

In particular, for all $a \in [0, 1)$

$$\lim_{n \rightarrow \infty} n^2 a^n = 0.$$

Thus for all $x \in (0, 1]$,

$$\lim_{n \rightarrow \infty} f_n(x) = x \lim_{n \rightarrow \infty} n^2 (1 - x^2)^n = 0$$

and obviously

$$\lim_{n \rightarrow \infty} f_n(0) = \lim_{n \rightarrow \infty} 0 = 0.$$

Therefore

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) = 0 \quad (x \in [0, 1])$$

and

$$\int_0^1 f(x) dx = \int_0^1 0 dx = 0.$$

On the other hand, for each n , by the change of variable $1 - x^2 \rightarrow x$,

$$\int_0^1 f_n(x) dx = n^2 \int_0^1 x(1 - x^2)^n dx = n^2 \int_0^1 x^n \frac{1}{2} dx = n^2 \frac{1}{2} \frac{1}{n+1} = \frac{n^2}{2n+2}.$$

Therefore

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \infty \neq 0 = \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx.$$

Due to the next example,

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx \neq \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx.$$

even though

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx < \infty.$$

Example 3.6. For all $n \in \mathbb{N}$, define

$$f_n(x) = nx(1 - x^2)^n. \quad (x \in \mathbb{R})$$

Then for each n , f_n is Riemann integrable on $[0, 1]$ since f is continuous on $[a, b]$. Observe that for $a \in [0, 1)$, by L'Hôpital's rule,

$$\lim_{x \rightarrow \infty} xa^x = \lim_{x \rightarrow \infty} \frac{x}{a^{-x}} = \lim_{x \rightarrow \infty} \frac{2}{-a^{-x} \ln a} = - \lim_{x \rightarrow \infty} a^x \frac{2}{\ln a} = 0.$$

and

$$\lim_{x \rightarrow \infty} x \cdot 0^x = 0.$$

In particular, for all $a \in [0, 1)$

$$\lim_{n \rightarrow \infty} na^n = 0.$$

Thus for all $x \in (0, 1]$,

$$\lim_{n \rightarrow \infty} f_n(x) = x \lim_{n \rightarrow \infty} n(1 - x^2)^n = 0$$

and obviously

$$\lim_{n \rightarrow \infty} f_n(0) = \lim_{n \rightarrow \infty} 0 = 0.$$

Therefore

$$f(x) := \lim_{n \rightarrow \infty} f_n(x) = 0 \quad (x \in [0, 1])$$

and

$$\int_0^1 f(x) dx = \int_0^1 0 dx = 0.$$

On the other hand, for each n

$$\int_0^1 f_n(x) dx = n \int_0^1 x(1 - x^2)^n dx = n \int_0^1 x^n \frac{1}{2} dx = n \frac{1}{2} \frac{1}{n+1} = \frac{n}{2n+2}.$$

Therefore

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \frac{1}{2} \neq 0 = \int_0^1 \lim_{n \rightarrow \infty} f_n(x) dx.$$

These examples lead us to find stronger convergence than pointwise convergence.

Definition 3.7. Let $E \subset \mathbb{R}$ and $\{f_n : n \in \mathbb{N}\}$ be a sequence of real-valued functions on E , i.e. for each $n \in \mathbb{N}$, f_n is a function from E to $\mathbb{R}$. We say that f_n converges to f uniformly on E if for any $\varepsilon > 0$, there exists a positive integer N such that for all $n \geq N$

$$\sup_{x \in E} |f_n(x) - f(x)| < \varepsilon.$$

Exercise 3.8. Let $E \subset \mathbb{R}$ and $\{f_n : n \in \mathbb{N}\}$ be a sequence of real-valued functions on E .

- (i) Show that if f_n converges to f uniformly on E , then f_n converges to f pointwise on E .
- (ii) Disprove that if f_n converges to f pointwise on E , then f_n converges to f uniformly on E .

Theorem 3.9. Let $E \subset \mathbb{R}$ and f_n be a sequence of real-valued functions on E . Then $f_n \rightarrow f$ converges to f uniformly on E if and only if for any $\varepsilon > 0$, there exists a positive integer N such that for all $n \geq N$ and $x \in E$,

$$|f_n(x) - f(x)| < \varepsilon.$$

Proof. We first prove the only if part ($\Rightarrow$). Let $\varepsilon > 0$. Then since $f_n \rightarrow f$ uniformly on E , there exists a $N \in \mathbb{N}$ such that for all $n \geq N$ such that

$$\sup_{x \in E} |f_n(x) - f(x)| < \varepsilon.$$

Since for all $n \geq N$ and $x \in E$,

$$|f_n(x) - f(x)| \leq \sup_{y \in E} |f_n(y) - f(y)|,$$

we have

$$|f_n(x) - f(x)| < \varepsilon.$$

Next we prove the if part ($\Leftarrow$). Let $\varepsilon > 0$. Then there exists a positive integer N such that for all $n \geq N$ and $x \in E$,

$$|f_n(x) - f(x)| < \varepsilon/2.$$

Thus for all $n \geq N$ we have

$$\sup_{x \in E} |f_n(x) - f(x)| \leq \varepsilon/2 < \varepsilon.$$

□

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